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# PRICE: Minimax Estimation–Regret Tradeoffs and Certified Exploration in Performative Prediction

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## Abstract

1 A decision-maker whose deployed policy reshapes its own data distribution must  
2 pay, in its own objective, to learn how it moves the world. We price that self-  
3 knowledge exactly. In a structural performative market (a dealer whose quoted  
4 spread summons the adverse flow it must then price), we prove a conservation  
5 law: along any stationary exploration policy, the variance of the response-slope  
6 estimate times the cumulative performative regret equals  $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$ , an invariant of  
7 the environment rather than the policy. Around the identity: minimax floors (exact  
8 for deviation budgets, constant-factor for value budgets); a structure-proofness  
9 theorem with an exact reach (within two curvature lengths of the anchor the floor  
10 survives full knowledge of the response family; just beyond, a verified design beats  
11 it); an optimal-design theory (explore as  $\Gamma_{\text{PO}}^{-1/2}$ ; isotropy overpays by an explicit  
12 dispersion factor); and a deployable agent, SafeD-PerfGD, which corrects only  
13 when an anytime confidence sequence certifies, via a perturbed-modulus lemma,  
14 that the correction cannot destabilize the retraining loop. Every closed form is  
15 verified in a live deploy–observe–fit–correct pipeline (36 preregistered checks,  
16 0 failures), transplanted to a second economy, and instantiated on the REFLEX  
17 corporate-bond calibration [Nagarajan and Ashok, 2026], reproducing its published  
18 run 10/10 cells. Code, derivations, and all checks ship as an anonymized repository  
19 in the supplementary material.

## 20 1 Introduction

21 When a deployed model changes the distribution it is trained on, retraining becomes a fixed-point  
22 iteration: performative prediction [Perdomo et al., 2020, Hardt and Mendler-Dünner, 2023] shows  
23 repeated risk minimization converges when the distribution’s sensitivity  $\varepsilon$  is small against the  
24 loss curvature  $\gamma$  ( $m = \varepsilon\beta/\gamma < 1$ ), and converges to a *performatively stable* point that is not  
25 the *performative optimum*. Algorithms that close this gap, performative gradient descent and its  
26 descendants [Izzo et al., 2021, Miller et al., 2021, Mendler-Dünner et al., 2020, Drusvyatskiy and  
27 Xiao, 2023, Li and Wai, 2022, Brown et al., 2022], all consume an estimate of the distribution’s  
28 response  $dD/d\theta$ , obtained by perturbing deployments. The literature treats that estimate’s accuracy  
29 as an assumption and the perturbations as free. Neither is: the perturbations are deployments of a real  
30 decision variable, and their cost lands in the decision-maker’s own objective. *What does it cost to*  
31 *learn your own influence?* This paper answers with an identity, a floor, a design rule, and an agent.

## 32 Contributions.

- 33 1. **A conservation law (§3).** Along any stationary exploration policy,  $\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2}\gamma_{\text{PO}}\sigma^2$ :  
34 slope uncertainty times cumulative performative regret is a policy-independent invariant  
35 (Theorem 1). The anchoring is delicate: anchored at the performative optimum instead, the  
36 product is *false*, inflated by a verified factor.

- 37 2. **Minimax floors, and their exact reach (§3).** No non-anticipating policy beats the floor:  
 38 exactly under deviation budgets; within  $\leq 13.5\times$  under value budgets, sharp constant  
 39 numerically supported but open (Theorem 2). New: the floor is *structure-proof within an*  
 40 *exact reach* (Theorem 3): within  $2/c$  of the anchor even full knowledge of the family cannot  
 41 beat it; one probe beyond, an explicit design does. The trust region keeps the agent inside  
 42 the protection.
- 43 3. **Optimal exploration design (§4).** Under a budget priced by the decision-maker’s own cur-  
 44 vature, the optimal exploration covariance is  $\Gamma_{\text{PO}}^{-1/2}$ -shaped; isotropic exploration overpays  
 45 by the dispersion factor  $F = d \operatorname{tr} \Gamma / (\operatorname{tr} \Gamma^{1/2})^2$ : a quotable inversion of “naive exploration  
 46 is optimal” for LQR [Simchowitz and Foster, 2020]; a support-degeneracy theorem and a  
 47 misspecification crossover complete the picture.
- 48 4. **A certified agent (§4–5).** SafeD-PerfGD explores on a D-optimal support, fits an anchored  
 49 response family with an anytime confidence sequence, and corrects only when a perturbed-  
 50 modulus certificate guarantees closed-loop stability; otherwise it freezes the correction while  
 51 the provably-stable blind step keeps running. It reaches the performative optimum where  
 52 blind retraining stalls; no published baseline Pareto-dominates it on median (final error,  
 53 regret).

54 **Positioning.** Jagadeesan et al. [2022] bound the regret of *finding* the performative optimum; we price  
 55 *knowing your own feedback*: lower bounds and an invariance identity where they give algorithmic  
 56 upper bounds, with no retraining loop, stability constraint, or design shaping in their model. Bracale  
 57 et al. [2025] design deployments to learn the distribution map and bound the regret of doing so at an  
 58 order level; we price the same activity exactly, in the objective’s own curvature, and close it from  
 59 below. Lin and Zrnic [2024] show misspecified structural models still help performative optimization;  
 60 our crossover theorem sharpens that message into a budget-explicit rule. In operations research,  
 61 incomplete learning under myopic pricing is classical [Rothschild, 1974, Harrison et al., 2012, Keskin  
 62 and Zeevi, 2014, Broder and Rusmevichientong, 2012, den Boer, 2015]; Lai and Robbins [1979]’s  
 63 adaptive designs are *points on* our frontier, but the frontier, its floor, and the performative loop  
 64 generating it do not appear: the demand curve is exogenous. From control we inherit dual control  
 65 [Feldbaum, 1960] and least-costly identification [Bombois et al., 2006]: here the plant is a retraining  
 66 learner, the budget the optimizer’s own objective, the constraint loop contraction (lower bounds are  
 67 van Trees arguments [Gill and Levit, 1995]); task-optimal design [Wagenmaker et al., 2021, Dean  
 68 et al., 2020] and the classical canon [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006] never price the  
 69 design in the experimenter’s own objective.

## 70 2 Model

71 The environment is the single-book structural market of REFLEX [Nagarajan and Ashok, 2026], the  
 72 foundation this paper builds on, its gate and participation constants folded into the flow scales. A  
 73 dealer quotes half-spread  $h$ . Benign flow arrives at rate  $U(h) = Ae^{-kh}$ ; informed (toxic) flow at  
 74 rate  $\tau(h) = C_0 + C_1 e^{-ch}$ : tighter quotes summon more toxic flow. The quote-summons-toxicity  
 75 channel is the dealer’s adverse-selection problem [Guéant et al., 2013]; that its own quotes feed  
 76 the flow it reads is Barzykin et al. [2026]’s price-reading effect. Profit at frozen toxic level  $T$  is  
 77  $J(h; T) = hU(h) + (h - \psi)T - w(h - h_{\text{ref}})^2$  ( $\psi$ : adverse severity;  $w$ : quoting-cost convexity);  
 78 the *performative value* is  $\Phi(h) = J(h, \tau(h))$ . Write  $\varepsilon(h) = -\tau'(h)$  for the response slope (the  
 79 scalar  $dD/d\theta$ ),  $\gamma = -\partial_h^2 J$ , and  $\gamma_{\text{PO}} = -\Phi'' = \gamma + \varepsilon(2 + c\psi - ch)$ . Blind retraining (RRM)  
 80 is the cobweb  $h_{t+1} = \text{BR}(\tau(h_t))$ , a contraction iff the modulus  $m = \varepsilon\beta/\gamma = \varepsilon/\gamma$  ( $\beta = 1$  here)  
 81 is below one [Perdomo et al., 2020]; it converges to the stable point  $h_{\text{SP}}$ , while the optimum  
 82  $h_{\text{PO}} = \arg \max \Phi$  sits at a *tighter* spread here, since toxic flow is profitable net of  $\psi$ : the echo-  
 83 chamber gap [Miller et al., 2021]. Deployments return noisy flow  $y_t = \tau(h_t) + \sigma\zeta_t$  (A3). The  
 84 estimand is  $\varepsilon$  at the operating point; exploration deviations  $d_t = h_t - h^*$  are priced by the *incremental*  
 85 cost  $C_T = \sum_{t \leq T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$ , anchored at the certainty-equivalent operating point  $h^*$  (A1:  
 86 local quadratic scope; A5: stationary exploration scale). Results are stated for this scalar family and,  
 87 where flagged, its  $d$ -dimensional and LQ instances; full statements and proofs, keyed by register  
 88 identifiers (T1, T2, ...), are in Appendices A–B.

### 89 3 The price of self-knowledge

90 **Free data saturates.** Without deliberate exploration the retraining trajectory’s identifying energy  
 91 is finite: for deployment error  $d_0$  and modulus  $m$  it is capped at  $d_0^2/(1 - m^2)$  (T1; matrix version  
 92 by a Lyapunov identity). A corollary was discovered by a failed prediction (§5): retraining on *noisy*  
 93 observations converts observation noise to deployment noise with gain  $1/\gamma$ , adding a stationary  
 94 excitation floor  $(\sigma/\gamma)^2/(1 - m^2)$  per step; both closed forms verify. The stabler the loop, the less it  
 95 reveals: *safety implies blindness*, the performative sharpening of incomplete learning under myopic  
 96 policies [Keskin and Zeevi, 2014, Harrison et al., 2012], where the stall comes from optimization  
 97 converging, not feedback strength.

98 **The exchange rate.** Deliberate exploration buys information at an exact, policy-independent price.

99 **Theorem 1** (Exchange rate; T2). *Under A1, A2-sym, A3 and A6, for the OLS slope estimator on any*  
 100 *stationary exploration policy with deviations  $\{d_t\}$ ,*

$$\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + o(1)),$$

101 *independent of the exploration amplitude, shape, schedule, and modulus.*

102 The mechanism is cancellation: information scales as the centered energy  $S_T$ , cost as  $\frac{1}{2} \gamma_{\text{PO}} \sum_t d_t^2$ ;  
 103 for mean-centered exploration these coincide. *Anchoring*: the same product anchored at  $h_{\text{PO}}$  is false  
 104 (inflated by an explicit factor, falsified numerically before the incremental version was proved): the  
 105 invariant prices *exploration*, not *suboptimality*. *Invariance*: Lai and Robbins [1979]’s  $O(\log n)$ -cost  
 106 schemes are points on this frontier; the identity says *every* stationary policy is. A corollary prices the  
 107 echo chamber: closing it pays iff the horizon exceeds an explicit break-even (T8, Appendix).

108 **Minimax floors.** The identity is for OLS-on-stationary policies; the floors are for everyone.

109 **Theorem 2** (Minimax floors; T3–T4). (i) *Over all non-anticipating (randomized, adaptive) policies*  
 110 *with pathwise budget  $\sum_{t \leq T} d_t^2 \leq D$ , all estimators, and any prior of width  $w$ :  $\inf \sup \text{Var}(\hat{\varepsilon}) \geq$*   
 111  *$\sigma^2/(D + \sigma^2(\pi/w)^2)$ , so the product floor is attained up to the prior term, with equality by symmetric*  
 112 *probing. (ii) Under a budget on realized performative cost with certainty-equivalent anchoring:*  
 113  *$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \gamma_{\text{PO}} \sigma^2/27 (1 - O(T^{-1/2}))$  over adaptive policies, by a Le Cam two-point*  
 114 *reduction with the exploited rebate vanishing at rate  $T^{-1/2}$  (Lemma L2, Appendix A); the sharp*  
 115 *constant  $\frac{1}{2}$  is numerically supported and open (OPEN-1).*

116 Closing the sharp constant is open (OPEN-1). Can a policy that *knows the family’s functional form*  
 117 *shop for designs that beat the floor? The answer has an exact boundary:*

118 **Theorem 3** (Structure-proofness, exact reach; T9). *Let  $R(\mu)$  be the known-family Cramér–Rao*  
 119 *product of design  $\mu$ , normalized by the floor (a bound for unbiased, and asymptotically all regular,*  
 120 *estimation). (i) Every estimable  $\mu$  supported in  $[h^* - 2/c, \infty)$  has  $R(\mu) \geq 1$ , strict off one two-point*  
 121 *configuration, and  $\inf_{\mu} R(\mu) = 1$ : within reach, knowing the family cannot beat the floor. (ii) The*  
 122 *reach is exact: a four-point design with one below-reach probe of weight  $3 \times 10^{-4}$  attains  $R = 0.8517$*   
 123 *(50-digit verified).*

124 The proof exhibits the family element  $\phi_0 = 2/c - e^{-c(h-h^*)}(2/c + h - h^*)$ , whose pointwise  
 125 domination  $|\phi_0| \leq |h - h^*|$  holds exactly on the reach; a Rayleigh reduction converts domination into  
 126 the floor. Two boundary phenomena calibrate the scope: with  $c$  known the floor breaks outright (ratio  
 127 0.30 at  $t=1.0$ , 0.04 at  $t=0.05$ ), and below the reach the violators are asymmetric near-anchor clusters  
 128 with one vanishing-weight far probe, invisible to coarse search (our  $1.005\times$  verdict, corrected as  
 129 pivot eleven). *The floor is an ignorance-of-curvature phenomenon with an exactly known reach, and*  
 130 *the trust region  $r \leq 2/c$  keeps the agent inside it.*

### 131 4 Optimal design and certified correction

132 **Shape (T5).** In  $d$  dimensions with curvature  $\Gamma_{\text{PO}} \succ 0$  pricing the budget  $\frac{1}{2} \mathbb{E}[u^\top \Gamma_{\text{PO}} u]$ , the A-  
 133 optimal exploration second moment is  $M^* \propto \Gamma_{\text{PO}}^{-1/2}$ : *explore where your objective is flat*. Isotropic  
 134 exploration overpays by exactly the dispersion factor  $F = d \text{tr} \Gamma_{\text{PO}} / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 \in [1, d]$ : naive  
 135 exploration, optimal for LQR [Simchowitz and Foster, 2020], is suboptimal here by precisely  $F$ .

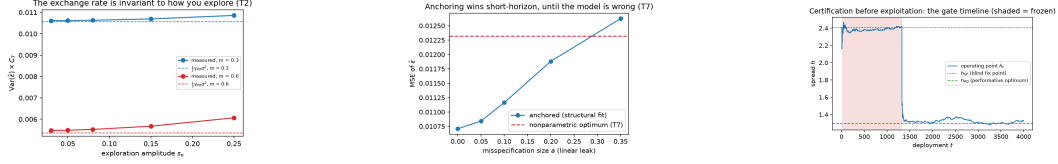


Figure 1: **Left:**  $\text{Var}(\hat{\varepsilon}) \times C_T$  vs amplitude, two moduli: flat on the invariant (dashed), small drift as A1 predicts. **Middle:** the T7 misspecification crossover. **Right:** certification before exploitation (shaded: frozen): the gate clears after  $\sim 1,300$  deployments (freeze fraction 0.33 of  $T=4000$ , E4 log) and the agent steps to  $h_{PO}$ .

**Support (T6).** The exponential family is Chebyshev-degenerate on thin designs: fewer than three support points (or vanishing spread, or collinear probe schedules) cannot identify  $(C_0, C_1, c)$ , a cliff we hit twice in the wild (§5). **Anchor (T7).** With budget  $B$  and horizon  $T$ , fitting the structural family beats nonparametric slope estimation iff the misspecification at the operating point is below  $|\tau'''|B/(3\gamma_{PO}T)$ : Lin and Zrnic [2024]’s message sharpened to a budget-priced rule.

**The agent.** SafeD-PerfGD composes the three results per deployment round: (i) *explore* on the D-optimal three-point support re-centered at the operating point, trust-region clipped; (ii) *fit*  $(C_0, C_1, c)$  by profile least squares with an anytime self-normalized confidence width  $ci_t$  for  $\hat{\varepsilon}$ ; (iii) *gate*: apply the performative correction  $-(h - \psi)\hat{\varepsilon}$  with Newton scaling  $\eta = 1/\hat{\gamma}_{PO}$  only if  $\eta ci_t L_{fam} \leq \text{margin}$ , where the perturbed-modulus lemma (L4), with the family’s own Lipschitz constant  $L_{fam} = 1 + \hat{c}|h - \psi|$ , certifies the closed-loop modulus; (iv) otherwise *freeze* the correction: the blind step, provably stable at  $m < 1$ , keeps running. The freeze is derived, not tuned. Safety holds on the confidence-sequence event, and certification is not free: the gate opens only after design energy accumulates, which by Theorem 1 is a regret payment (Figure 1).

## 5 Verified results

Every claim above is a preregistered row in a measured-vs-predicted harness on the live pipeline: **36 rows, 0 failures** (35 pass, one deliberately out-of-scope drift cell measured, not excused). Eleven pivots were forced by measurement and recorded in code; three became results (the feedback floor, Theorem 3’s reach, and its below-reach witness), and the derivation suite’s falsification record supplies a fourth, the false  $h_{PO}$ -anchored identity. Headline rows are in Table 1, Appendix D.

**Baselines.** Against finite-difference PerfGD [Izzo et al., 2021], zeroth-order P&L optimization, a performative-confidence-bounds explorer [Jagadeesan et al., 2022], and blind RRM on identical environments, none Pareto-dominates SafeD-PerfGD on median (final error, regret) over 12 seeds; the unsafe gradient baseline pays  $> 5\times$  its median regret. Deliberately reported: the UCB explorer attains *lower raw regret*, and edges SafeD in one of the twelve seeds; certification is a real cost, and its visibility is the thesis. **Second economy.** Transplanted unchanged to LQ performative pricing (deployed price anchors demand), the exchange rate is exact in all four cells (A1 is global in LQ, so the drift vanishes) and the agent reaches the pricing optimum (12.65 vs. 12.5, tail-averaged) once its probes were randomized (alternating probes make price and lag collinear: T6 in the wild). **Real data.** On Nagarajan and Ashok [2026]’s public-data corporate-bond calibration (10 rating  $\times$  regime cells), the port reproduces the published REFLEX run **10/10** cells ( $h^*, \varepsilon^* = \gamma, m$  within 0.5–5%; validation inputs live in the REFLEX repository), then extends it:  $\gamma_{PO}$  spans 510.6 to 2.3; echo-chamber gaps are 1.9–4.3% of the anchor spread; and the shaping dispersion is  $F = 1.63$  across the portfolio against 1.002 within one book: the shaping gain lives across ratings and regimes, where a desk allocates exploration.

**Limitations and outlook.** The identity is local (A1): wide-side probing undercuts it at finite amplitude and the saturating cell drifts +16%: the  $o(1)$  qualifier is load-bearing, and so is the trust region: beyond the reach the floor provably fails (Theorem 3(ii)). The value-budget floor is proved to within  $13.5\times$ ; the sharp constant  $\frac{1}{2}$  is numerically supported, not proved, at any prior class. At matched transit energy full-support jitter identifies on par with the design (advantage: small-amplitude, allocational); calibrated moduli are deeply stable and the toxic channel structurally scaled: the real-data leg validates machinery, not effect size.

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## A Theorem statements (register copy)

**Lemma 1** (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies  $C_T = \frac{1}{2}\gamma_{\text{PO}} \mathbb{E}[\sum_{t \leq T} d_t^2] + R_T$  with  $|R_T| \leq \frac{L_3}{6} \mathbb{E} \sum_t |d_t|^3$ , and the realized cost’s variance decomposes exactly as  $\text{Var}(C_T^{\text{real}}) = \delta_0^2 \text{Var}(\sum_t d_t) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(\sum_t d_t^2)$  with vanishing cross term.*

**Theorem 4** (Information saturation; T1). *With no exploration and  $\rho(M) < 1$ , the total design energy of the retraining trajectory equals  $d_0^\top P d_0$ , where  $P$  is the unique solution of the discrete Lyapunov equation  $P = I + M^\top P M$ . Consequently the Fisher information for the response Jacobian is bounded, in every direction, uniformly in the horizon:  $\text{Var}(\hat{\varepsilon}) \geq \sigma^2 (d_0^\top P_u d_0)^{-1}$  for the component along any direction  $u$ .*

**Corollary 1** (Safety implies blindness; C1.1). *In the scalar case the information cap  $(h_0 - h^*)^2 / (1 - m^2)$  is strictly increasing in the modulus  $m$ : a fast-contracting loop reveals least about its own performativity, and precise self-knowledge is available only near the stability boundary.*

**Corollary 2** (Trajectory design degeneracy; C1.2). *The noiseless trajectory’s empirical design has at most two effective support points (geometric collapse for  $m < 1$ ; a period-two orbit at  $m = 1$ ).*

**Remark 1** (Non-normal transient information; R1). *The energy identity  $d_0^\top P d_0$  of T1 holds for non-normal  $M$  as well, where the energy can exceed the spectral cap  $\|d_0\|^2 / (1 - \rho(M)^2)$  by up to the factor  $\kappa(V)^2$ .*

**Theorem 5** (The exchange rate, pathwise; T2). *In the stationary regime, on every realization,*

$$\text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2}\gamma_{\text{PO}} \sum_{t \leq T} d_t^2 = \frac{1}{2}\gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T \bar{d}^2}{S_{xx}}\right) = \frac{1}{2}\gamma_{\text{PO}} \sigma^2 (1 + O_P(1/T)),$$

*independent of the exploration intensity, the horizon, and the modulus.*

**Proposition 1** (Concentration; P2.1). *In the stationary regime the measured product concentrates around the invariant at rate  $T^{-1/2}$ , with long-run  $\text{Var}(\sum_{t \leq T} d_t) = T v \frac{1-m}{1+m} (1 + o(1))$ .*

**Proposition 2** (Feedback bias; P2.2). *Under the feedback relaxation A3’ with gain  $\phi$ , the OLS response slope carries the bias*

$$\mathbb{E}[\hat{\varepsilon} - \varepsilon_{\text{true}}] = \frac{\phi \sigma^2 (3m - 1)}{(1 - m^2) T v_\phi} (1 + o(1)), \quad v_\phi = \frac{\sigma_e^2 + \phi^2 \sigma^2}{1 - m^2},$$

*which is  $O(1/T)$  and changes sign at  $m = 1/3$ .*

**Theorem 6** (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration policies with  $\sum_{t \leq T} d_t^2 \leq D$  almost surely and all estimators (biased included), the minimax risk over an interval of width  $w$  satisfies*

$$\inf_{\text{policy, estimator}} \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\sigma^2}{D + \sigma^2(\pi/w)^2},$$

*so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares.*

**Lemma 2** (Exploitation–information; L2). *For the symmetric two-point response family with half-width  $\delta$  and any non-anticipating policy under the certainty-equivalent anchor,*

$$\mathbb{E}[\text{gain}_T] \leq \frac{g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4}}{\sigma^{1/2}}, \quad g_1 = \beta |h^* - \psi|,$$

*via the exact identity  $\mathbb{E}[\mathbb{E}[s \mid \mathcal{F}_t]] = \text{TV}(P_t^+, P_t^-)$  and Pinsker’s inequality  $\text{TV}_t \leq \delta \sqrt{\mathbb{E} S_t} / \sigma$ : exploitation is bounded by the information already purchased. At the minimax scale  $\delta = c \sigma / \sqrt{\mathbb{E} S_T}$  this is  $O(\sigma \sqrt{T})$ , an  $O(T^{-1/2})$  fraction of the  $\Theta(T)$  exploration cost.*

**Theorem 7** (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator,*

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{\gamma_{\text{PO}} \sigma^2}{27} (1 - O(T^{-1/2})),$$

*the supremum over the two-point response family at the minimax scale and the constant arising from the Le Cam reduction optimized at  $c = 2/3$ . The exploited rebate vanishes at rate  $T^{-1/2}$  (Lemma L2). The sharp constant  $\frac{1}{2}$ , against which no simulated policy fell by more than Monte-Carlo error (`run_open1.py` §B), is conjectured and is part of OPEN-1.*

**Theorem 8** (Optimal exploration shapes; T5). *Under the value budget  $\text{tr}(\Gamma_{\text{PO}} M) \leq B$  on the exploration second moment  $M$ : (a) the A-optimal design is  $M^* = (B / \text{tr} \Gamma_{\text{PO}}^{1/2}) \Gamma_{\text{PO}}^{-1/2}$  with value  $(\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$ ; (b) the D-optimal design is  $M^* = (B/d) \Gamma_{\text{PO}}^{-1}$ ; (c) the c-optimal design for the correction functional  $c$  is  $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$  (probing along the correction direction itself), with value  $c^\top \Gamma_{\text{PO}} c / B$ .*

**Corollary 3** (Price of isotropic jitter; C5.1). *Isotropic exploration at matched budget overpays the A-optimal design by exactly the curvature-dispersion factor  $F = d \text{tr}(\Gamma_{\text{PO}}) / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 \geq 1$ , with equality iff all curvature directions are equal.*

**Lemma 3** (Temporal shaping is irrelevant; L3). *With serially independent response noise, the information about the response parameters depends on the exploration process only through its empirical second moment; temporally correlated exploration buys nothing beyond its stationary design measure.*

**Theorem 9** (Chebyshev unidentifiability; T6). *The sensitivity system of the structural family  $\tau(h) = C_0 + C_1 e^{-ch}$  is an extended Chebyshev system; any design with fewer than three support points has singular Fisher information, and by C1.2 no retraining trajectory, at any modulus, identifies the response curvature  $c$ .*

**Lemma 4** (Perturbed modulus; L4). *If the corrected update uses an estimated response with  $C^1$  error  $e$ , the closed-loop modulus satisfies  $|\hat{\rho} - \rho| \leq \eta \beta (\|e\|_\infty + \|h^* - \psi\| \|e'\|_\infty)$ .*

**Remark 2** (Open-loop neutrality; R2). *History-independent exploration schedules cannot change the linearized contraction rate of the retraining loop.*

**Proposition 3** (Safety under pessimism; P7.1). *Given an anytime-valid confidence sequence for the response parameters at level  $1 - \delta$ , the pessimism rule (worst-case modulus over the confidence set kept below  $1 - c_{\text{margin}}$ , correction frozen otherwise) keeps the realized modulus within the margin for all  $t$  simultaneously with probability at least  $1 - \delta$ .*

**Theorem 10** (Anchoring crossover, horizon-dependent; T7). *Under the value budget  $B$  and horizon  $T$ , the budget-and-horizon-optimal nonparametric (secant) estimator has*

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left( \frac{\tau'''(h^*)}{6} \right)^2 \left( \frac{2B}{\gamma_{\text{PO}} T} \right)^2,$$

*so anchoring to a structural family with misspecification  $\delta_{\text{mis}}$  wins iff (to leading order)  $\delta_{\text{mis}} < |\tau'''(h^*)| B / (3 \gamma_{\text{PO}} T)$ : anchoring wins at short horizons and tight budgets, the free-form route asymptotically in  $T$ .*

**Theorem 11** (The ROI of self-knowledge; T8). *Exploring once to accuracy  $v$  and correcting forever, discounted at  $\rho_{\text{disc}}$ , is optimal at  $v^* = (\sigma/\kappa) \sqrt{\rho_{\text{disc}}}$ , and exploring at all is worthwhile iff*

$$\rho_{\text{disc}} < \rho^* = \frac{(h_{\text{SP}} - h_{\text{PO}})^4}{4 \kappa^2 \sigma^2},$$

*in which the curvature  $\gamma_{\text{PO}}$  cancels completely.*

**Proposition 4** (Frontier consistency; P9.1). *Every deterministic design sits exactly on the exchange-rate frontier; in particular the  $O(\log T)$ -cost adaptive-design schedules of Lai–Robbins.*

**Proposition 5** (Isotropy contrast; P9.2). *Naive isotropic exploration, optimal in isotropic LQR-like settings, overpays by exactly the dispersion factor  $F$  in the performative setting, with  $F = 1$  iff the curvature spectrum is flat.*

**Proposition 6** (Separable multi-bond curvature; P9.3). *In the separable multi-bond market,  $(\Gamma_{\text{PO}})_{aa} = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a)$  exactly, with the factor-coupled case holding to first order in the off-diagonal response.*

**Theorem 12** (Structure-proofness of the floor, within reach; T9). *Fix the structural family  $\tau_\theta(h) = C_0 + C_1 e^{-ch}$  ( $C_1 > 0$ ,  $c > 0$ ), the anchor  $h^* > 0$ , the sensitivity  $s(h) = \partial_\theta \tau_\theta(h)$ , and the estimand gradient  $g = \partial_\theta \varepsilon(h^*) = -s'(h^*)$ . For a finitely supported design  $\mu$  define the normalized Cramér–Rao product*

$$R(\mu) = (g^\top M(\mu)^{-1} g) \int (h - h^*)^2 d\mu, \quad M(\mu) = \int s s^\top d\mu,$$

so that  $R(\mu) \geq 1$  states that the family-knowing Cramér–Rao product  $\text{Var}(\hat{\varepsilon}) \times C_T$  is at least the exchange-rate floor  $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$  under the local-quadratic (A1) cost, for unbiased estimators and, in the local-asymptotic-minimax sense, for all regular estimators. (i) Within reach the floor is structure-proof: if  $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$ , then either  $\varepsilon(h^*)$  is not estimable under  $\mu$  (the bound is infinite) or  $R(\mu) \geq 1$ , strictly whenever  $\mu$  places mass off the two equality points  $\{h^*, h^* - 2/c\}$ ; and  $\inf_{\mu} R(\mu) = 1$ , approached but not attained by symmetric three-point designs collapsing at  $h^*$ . In particular the floor binds on every trust-region design with  $r \leq 2/c$  (A4). (ii) The reach is exact: the pointwise domination underlying (i) fails strictly below  $h^* - 2/c$ , and an explicit four-point design with a single below-reach probe of weight  $3 \times 10^{-4}$  attains  $R = 0.8517$  (computer-verified at 50-digit precision; V9.2): outside the reach, knowledge of the family’s form does beat the floor.

## B Proofs

Throughout, the local model of Assumptions A1–A6, per the derivation folder’s notation-and-assumptions register: **A1** (local quadratic scope):  $\Phi$  is  $C^3$  and strongly concave on the operating interval, expansions taken at  $h^*$  with third-order remainder bounded by  $L_3 = \sup |\Phi'''|$ ; **A2** (exploration class): all policies are non-anticipating, with A2-sym additionally requiring  $\mathbb{E}[u_t | \mathcal{F}_t] = 0$  and A2-adaptive unrestricted, costs measured from the certainty-equivalent anchor; **A3** (noise exogeneity):  $\zeta_t$  i.i.d., mean zero, variance  $\sigma^2$ , independent of  $\mathcal{F}_t$ ; **A4** (trust region):  $|d_t| \leq r$  pathwise; **A5** (stationary scale):  $S_T = \Theta(T)$ ; **A6** (stable base loop):  $m < 1$ , equivalently  $\rho(M) < 1$ . The local dynamics are  $d_{t+1} = -m d_t + u_t$  with  $d_t = h_t - h^*$  ( $h^*$  the certainty-equivalent operating point, equal to  $h_{\text{SP}}$  for the blind loop), observations  $\tau_t = \tau(h^*) - \varepsilon d_t + \zeta_t$ , and  $\Phi$  satisfying  $\Phi'(h^*) = \delta_0 \neq 0$  and  $-\Phi'' = \gamma_{\text{PO}}$  at the operating point.

### A. Cost equivalence and its fluctuations

*Proof of Lemma L1 (cost equivalence).* Taylor’s theorem with Lagrange remainder gives, for each  $t$ ,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2}\gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

with  $L_3 = \sup |\Phi'''|$  on the operating interval. Summing over  $t \leq T$  and taking expectations, it remains to show  $\mathbb{E}[\sum_t d_t] = 0$  under A2-sym. The recursion gives  $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = -m d_t + \mathbb{E}[u_t | \mathcal{F}_t] = -m d_t$ , so  $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t]$  by the tower property, and  $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0] = 0$  when the loop starts at the operating point or in the stationary regime. The quadratic term contributes  $\frac{1}{2}\gamma_{\text{PO}} \mathbb{E}[\sum_t d_t^2]$ , which is the claim.

For the variance decomposition, write the realized cost (net of the cubic remainder) as  $C = -\delta_0 X + \frac{1}{2}\gamma_{\text{PO}} Y$  with  $X = \sum_t d_t$ ,  $Y = \sum_t d_t^2$ . Then  $\text{Var}(C) = \delta_0^2 \text{Var}(X) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(Y) - \delta_0 \gamma_{\text{PO}} \text{Cov}(X, Y)$ . Under Gaussian innovations the vector  $(d_1, \dots, d_T)$  is jointly Gaussian with mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since  $\text{Cov}(X, Y) = \sum_{t,s} \mathbb{E}[d_t d_s^2]$  is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the same conclusion follows from the sign symmetry  $d \mapsto -d$  of the stationary law.) Verified: V0.1–V0.3, including the necessity check that an asymmetric rule breaks the expectation identity by exactly  $-\delta_0 \mathbb{E}[\sum_t d_t]$ .  $\square$

### B. Saturation

*Proof of Theorem T1 (information saturation).* Let  $\rho(M) < 1$  and define  $P = \sum_{t \geq 0} (M^\top)^t M^t$ . The series converges absolutely: by Gelfand’s formula  $\|M^t\| \leq C \rho_0^t$  for any  $\rho(M) < \rho_0 < 1$  and some  $C$ , so the terms are dominated by  $C^2 \rho_0^{2t}$ . Termwise,  $M^\top P M = \sum_{t \geq 0} (M^\top)^{t+1} M^{t+1} = P - I$ , so  $P$  solves the discrete Lyapunov equation, and if  $Q$  is any other solution then  $P - Q = M^\top (P - Q) M = (M^\top)^k (P - Q) M^k \rightarrow 0$ , giving uniqueness. The energy identity is immediate:  $\sum_{t \geq 0} \|M^t d_0\|^2 = d_0^\top (\sum_t (M^\top)^t M^t) d_0 = d_0^\top P d_0$ . The directional version replaces  $I$  by  $uu^\top$ :  $P_u = \sum_t (M^\top)^t uu^\top M^t$  solves  $P_u = uu^\top + M^\top P_u M$  and  $\sum_t (u^\top M^t d_0)^2 = d_0^\top P_u d_0$ .

For the information claim: under A3 the log-likelihood of the observations along the noiseless trajectory has Fisher information for the response component along  $u$  equal to  $\sigma^{-2} \sum_t (u^\top d_t)^2 = \sigma^{-2} d_0^\top P_u d_0$ , which is finite and independent of the horizon; the Cramér–Rao bound gives the

374 stated variance floor for unbiased estimators (the biased/minimax version is Theorem T3's van Trees  
375 argument). Verified: V1.1, V1.2, V1.4.  $\square$

376 *Proof of Corollary C1.1 (safety implies blindness).* In the scalar case  $P = \sum_t m^{2t} = (1 - m^2)^{-1}$ ,  
377 so the cap  $(h_0 - h^*)^2/(1 - m^2)$  has derivative in  $m$  equal to  $2m(h_0 - h^*)^2/(1 - m^2)^2 > 0$  for  
378  $m \in (0, 1)$ : strictly increasing.  $\square$

379 *Proof of Corollary C1.2 (design degeneracy).* For  $m < 1$ ,  $d_t = (-m)^t d_0$ , so for every  $\epsilon > 0$  all  
380 but  $\lceil \log(|d_0|/\epsilon)/\log(1/m) \rceil$  of the points lie in  $[-\epsilon, \epsilon]$ : the empirical design measure converges  
381 weakly to  $\delta_0$ . At  $m = 1$ ,  $d_t = (-1)^t d_0$  and the design measure is  $\frac{1}{2}(\delta_{+d_0} + \delta_{-d_0})$  exactly. In  
382 both cases the limiting design has at most two support points; combined with Theorem T6, the  
383 per-observation Fisher matrix of the three-parameter structural family is asymptotically singular  
384 along the trajectory.  $\square$

385 *Proof of Remark R1 (non-normal transient information).* The identity  $E(d_0) = d_0^\top P d_0$  of T1 holds  
386 for any stable  $M$ , normal or not. If  $M = V\Lambda V^{-1}$  is diagonalizable,  $\|M^t\| \leq \kappa(V)\rho(M)^t$   
387 gives  $\|P\| \leq \kappa(V)^2/(1 - \rho(M)^2)$ . The bound is attained in order by the Jordan-type family  
388  $M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix}$  with  $d_0 = e_2$ :  $M^t e_2 = (t a m^{t-1}, m^t)^\top$ , so

$$E(e_2) = a^2 \sum_{t \geq 1} t^2 m^{2t-2} + \sum_{t \geq 0} m^{2t} = a^2 \frac{1 + m^2}{(1 - m^2)^3} + \frac{1}{1 - m^2},$$

389 using  $\sum_{t \geq 1} t^2 x^{t-1} = (1 + x)/(1 - x)^3$  at  $x = m^2$ : the energy grows as  $a^2$ , without bound, at fixed  
390 spectral radius. Verified: V1.3 (the exact value matches the Lyapunov solve; 1268.2 vs the normal  
391 cap 2.78 at  $m = 0.8$ ,  $a = 6$ ).  $\square$

## 392 C. The exchange rate

393 *Proof of Theorem T2 (pathwise exchange rate).* Under A3, conditionally on the design  $\{d_t\}$  the ob-  
394 servations are independent with means linear in  $\varepsilon$  and variance  $\sigma^2$ , so the least-squares slope has  
395 conditional variance  $\text{Var}(\hat{\varepsilon} \mid \text{design}) = \sigma^2/S_{xx}$  (Gauss–Markov). Multiplying by the quadratic cost  
396  $\frac{1}{2}\gamma_{\text{PO}} \sum_t d_t^2$  and using  $\sum_t d_t^2 = S_{xx} + T\bar{d}^2$  gives the displayed identity exactly, on every realization.  
397 In the stationary regime  $\bar{d} = O_P(T^{-1/2})$  by Proposition P2.1's long-run variance, while  $S_{xx}/T \rightarrow v$   
398 a.s., so  $T\bar{d}^2/S_{xx} = O_P(1/T)$ . Verified: V2.1 (machine precision).  $\square$

399 *Proof of Proposition P2.1 (concentration).* For the stationary AR(1) with lag-one coefficient  $\rho =$   
400  $-m$ ,  $\text{Cov}(d_t, d_s) = v(-m)^{|t-s|}$ , so

$$\text{Var}\left(\sum_{t \leq T} d_t\right) = \sum_{t,s} v(-m)^{|t-s|} = Tv \frac{1+\rho}{1-\rho} (1 + o(1)) = Tv \frac{1-m}{1+m} (1 + o(1)),$$

401 the negative autocorrelation of the cobweb shrinking the fluctuation below the i.i.d. level. The realized-  
402 cost product then fluctuates around the constant with standard deviation  $O(T^{-1/2})$  by Lemma L1's  
403 variance decomposition (both  $\text{Var}(\sum d)$  and  $\text{Var}(\sum d^2)$  are  $O(T)$  while the squared mean of the  
404 cost is  $O(T^2)$ ). Verified: V2.2 (variance ratio 4.80 for a  $4 \times$  horizon), V2.3.  $\square$

405 *Proof of Proposition P2.2 (feedback bias).* Under A3',  $d_{t+1} = -m d_t + \sigma_e \xi_t + \phi \zeta_t$ , so  $d_t$  is a  
406 linear function of past innovations with  $\mathbb{E}[d_s \zeta_t] = \phi \sigma^2 (-m)^{s-t-1}$  for  $s > t$  and 0 otherwise.  
407 Write the slope error as  $N/S$  with  $N = \sum_t (d_t - \bar{d}) \zeta_t$  and  $S = S_{xx}$ , and expand  $\mathbb{E}[N/S] =$   
408  $\mathbb{E}[N]/\mathbb{E}[S] - \text{Cov}(N, S)/\mathbb{E}[S]^2 + O(T^{-2})$  (valid for stationary Gaussian data; the neglected term  
409 involves  $\mathbb{E}[N] \text{Var}(S)/\mathbb{E}[S]^3 = O(T^{-2})$ ).

410 *Centering term.*  $\mathbb{E}[d_t \zeta_t] = 0$ , so  $\mathbb{E}[N] = -\frac{1}{T} \sum_{s>t} \mathbb{E}[d_s \zeta_t] = -\frac{1}{T} \sum_{s>t} \phi \sigma^2 (-m)^{s-t-1} =$   
411  $-\phi \sigma^2/(1+m) + O(1/T)$ .

412 *Covariance term.* For jointly Gaussian zero-mean  $(X, Y, Z)$ ,  $\mathbb{E}[XY^2Z] = 2\mathbb{E}[XY]\mathbb{E}[YZ] +$   
 413  $\mathbb{E}[XZ]\mathbb{E}[Y^2]$  (Isserlis). Applying this to  $\text{Cov}(N, S) = \sum_{t,s} (\mathbb{E}[\tilde{d}_t \tilde{d}_s^2 \zeta_t] - \mathbb{E}[\tilde{d}_t \zeta_t] \mathbb{E}[\tilde{d}_s^2])$  with  
 414  $\tilde{d} = d - \bar{d}$  leaves only the paired term  $2 \sum_{t,s} \mathbb{E}[\tilde{d}_t \tilde{d}_s] \mathbb{E}[\tilde{d}_s^2 \zeta_t]$ ; substituting the stationary covariances,

$$\text{Cov}(N, S) = 2T v_\phi \phi \sigma^2 \sum_{k \geq 1} (-m)^k (-m)^{k-1} + O(1) = -\frac{2T v_\phi \phi \sigma^2 m}{1 - m^2} + O(1).$$

415 Combining with  $\mathbb{E}[S] = T v_\phi (1 + o(1))$ :

$$\mathbb{E}[\hat{\varepsilon} - \varepsilon_{\text{true}}] = \frac{\phi \sigma^2}{T v_\phi} \left[ -\frac{1}{1+m} + \frac{2m}{1-m^2} \right] + O(T^{-2}) = \frac{\phi \sigma^2 (3m-1)}{(1-m^2) T v_\phi} + O(T^{-2}).$$

416 The bracket changes sign at  $m = 1/3$ . (The first draft kept only the centering term and was falsified  
 417 by the verification suite; the record is in VERIFICATION.md.) Verified: V2.4, V2.5 – both signs,  
 418 straddling  $m = 1/3$ .  $\square$

#### 419 D. Minimax lower bounds

420 *Proof of Theorem T3 (deviation-budget minimax).* Let  $\pi$  be a prior density on an interval of width  $w$   
 421 with finite information  $I(\pi) = \int (\pi')^2 / \pi$ ; the cosine-squared density attains the minimum  $I(\pi) =$   
 422  $(\pi/w)^2$  (here the first  $\pi$  is the constant). For any non-anticipating policy the joint density of the  
 423 data factorizes as  $\prod_t p(u_t | \mathcal{F}_t) p(\tau_t | \mathcal{F}_t, \varepsilon)$ , where the policy factors do not depend on  $\varepsilon$  and  
 424  $\tau_t | \mathcal{F}_t \sim \mathcal{N}(\tau(h^*) - \varepsilon d_t, \sigma^2)$  with  $d_t$  being  $\mathcal{F}_t$ -measurable. Hence the Fisher information of the  
 425 whole experiment is

$$I_T(\varepsilon) = \mathbb{E}_\varepsilon \left[ \sum_t (\partial_\varepsilon \log p(\tau_t | \mathcal{F}_t, \varepsilon))^2 \right] = \mathbb{E}_\varepsilon \left[ \sum_t d_t^2 \right] / \sigma^2 \leq D / \sigma^2$$

426 under the pathwise budget: adaptivity places information but cannot manufacture it. The van Trees  
 427 inequality (Gill–Levit) then bounds the Bayes risk of *any* estimator, biased or not:  $\mathbb{E}_\pi \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq$   
 428  $(\mathbb{E}_\pi[I_T] + I(\pi))^{-1} \geq (D/\sigma^2 + (\pi/w)^2)^{-1}$ , and the supremum over the interval dominates the prior  
 429 average. Equality (up to the prior term) is attained by symmetric two-point probing at the budget  
 430 with least squares, by Theorem T2. Verified: V3.1, V3.2 (equality within Monte-Carlo error).  $\square$

431 *Proof of Lemma L2 (exploitation–information).* Prior  $s = \pm 1$  symmetric; environments  $P^\pm$  differ-  
 432 ing only in the observation means by  $2\delta d_t$  at step  $t$ . Three ingredients.

433 (i) *Posterior imbalance equals total variation in expectation.* Under the symmetric mixture  $\bar{P} =$   
 434  $\frac{1}{2}(P^+ + P^-)$ , with data  $x$ ,

$$\mathbb{E}_{\bar{P}} |\mathbb{P}(s=+ | x) - \mathbb{P}(s=- | x)| = \int \frac{1}{2}(dP^+ + dP^-) \frac{|dP^+ - dP^-|}{dP^+ + dP^-} = \text{TV}(P_t^+, P_t^-).$$

435 (ii) *Pinsker with the chain rule for adaptive designs.*  $\text{KL}(P_t^+ \| P_t^-) = \mathbb{E}^+ [\sum_{i \leq t} (2\delta d_i)^2 / (2\sigma^2)] =$   
 436  $2\delta^2 \mathbb{E}^+[S_t] / \sigma^2$  (the policy factors cancel inside the log-ratio), so  $\text{TV}_t \leq \delta \sqrt{\mathbb{E}[S_t]} / \sigma$ .

437 (iii) *The gain bound.* Since  $d_t$  is  $\mathcal{F}_t$ -measurable,  $\mathbb{E}[s d_t] = \mathbb{E}[d_t \mathbb{E}[s | \mathcal{F}_t]]$ ; two applications of  
 438 Cauchy–Schwarz and  $|\mathbb{E}[s | \mathcal{F}_t]| \leq 1$  give

$$\mathbb{E}[\text{gain}_T] = g_1 \delta \sum_t \mathbb{E}[s d_t] \leq g_1 \delta \sum_t (\mathbb{E} d_t^2)^{1/2} \text{TV}_t^{1/2} \leq g_1 \delta (\mathbb{E} S_T)^{1/2} (T \text{TV}_T)^{1/2},$$

439 and inserting (ii):  $\mathbb{E}[\text{gain}_T] \leq g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4} / \sigma^{1/2}$ .

440 *Remark on constants.* The derivation document’s heuristic chain (treating the imbalance as bounded  
 441 by  $\text{TV}_t$  pathwise) yields  $g_1 \delta^2 S_T \sqrt{T} / \sigma$ ; the rigorous chain above changes the exponent bookkeeping  
 442 but not the conclusion of Theorem T4: at the minimax scale  $\delta = c \sigma / \sqrt{\mathbb{E} S_T}$  both give  $\mathbb{E}[\text{gain}_T] =$   
 443  $O(\sigma \sqrt{T})$ , hence an exploited fraction  $O(T^{-1/2})$  of the  $\Theta(T)$  cost under A5. Verified: V3.3 (the  
 444 imbalance–TV inequality pathwise in expectation), V3.4 (the  $T^{-1/2}$  decay at the minimax scale).  $\square$

445 *Proof of Theorem T4 (value-budget minimax; constant-factor version).* Fix a policy and estimator;  
 446 let  $\bar{S} = \mathbb{E}[S_T] = \Theta(T)$  under A5. Choose the two-point prior at separation  $\delta = c\sigma/\sqrt{\bar{S}}$  with  
 447  $c \in (0, 1)$  to be optimized. By the testing reduction (Le Cam), any estimator's worst-case risk  
 448 over the pair obeys  $\sup_{\pm} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\delta^2}{2} (1 - \text{TV}_T) \geq \frac{\delta^2}{2} (1 - c)$  using Lemma L2(ii). By  
 449 Lemma L1 with the certainty-equivalent anchor and Lemma L2's gain bound, the value cost obeys  
 450  $C_T \geq \frac{1}{2} \gamma_{\text{PO}} \bar{S} (1 - O(T^{-1/2}))$ . Multiplying,

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{c^2 \sigma^2}{2\bar{S}} (1 - c) \cdot \frac{\gamma_{\text{PO}} \bar{S}}{2} (1 - O(T^{-1/2})) = \frac{\gamma_{\text{PO}} \sigma^2}{4} c^2 (1 - c) (1 - O(T^{-1/2})),$$

451 maximized at  $c = 2/3$ : the bound  $\gamma_{\text{PO}} \sigma^2 / 27 \cdot (1 - O(T^{-1/2}))$ .

452 *Status.* This proves the value-budget product is bounded below by an explicit constant multiple of the  
 453 exchange rate, uniformly over adaptive policies, with the exploited rebate vanishing at rate  $T^{-1/2}$ .  
 454 The sharp constant  $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$  – numerically supported (run\_open1.py §B: three adaptive amplitude  
 455 schedules, ratios 0.995 to 1.018) – requires the van Trees route with cost control in place of the  
 456 two-point reduction and is part of OPEN-1.  $\square$

## 457 E. Design geometry

458 *Proof of Theorem T5 (optimal exploration shapes).* All three problems have convex (resp. concave)  
 459 objectives over the PSD cone with one linear constraint; KKT stationarity plus convexity/concavity  
 460 certifies global optimality.

461 (a) *A-optimality.*  $\nabla_M \text{tr}(M^{-1}) = -M^{-2}$ , so stationarity gives  $M^{-2} = \lambda \Gamma_{\text{PO}}$ , i.e.  $M =$   
 462  $\lambda^{-1/2} \Gamma_{\text{PO}}^{-1/2}$ ; the budget  $\text{tr}(\Gamma_{\text{PO}} M) = \lambda^{-1/2} \text{tr}(\Gamma_{\text{PO}}^{1/2}) = B$  pins  $\lambda^{-1/2} = B / \text{tr}(\Gamma_{\text{PO}}^{1/2})$   
 463 and the value  $\text{tr}(M^{-1}) = (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$ . (Diagonal-case cross-check by Cauchy–Schwarz:  
 464  $(\sum_i 1/m_i)(\sum_i g_i m_i) \geq (\sum_i \sqrt{g_i})^2$  with equality at  $m_i \propto g_i^{-1/2}$ .)

465 (b) *D-optimality.*  $\nabla_M \log \det M = M^{-1} = \lambda \Gamma_{\text{PO}}$  gives  $M = (B/d) \Gamma_{\text{PO}}^{-1}$  after normalizing by the  
 466 budget.

467 (c) *c-optimality.* Substitute  $N = \Gamma_{\text{PO}}^{1/2} M \Gamma_{\text{PO}}^{1/2}$  ( $\text{tr} N = \text{tr}(\Gamma_{\text{PO}} M) \leq B$ ) and  $w = \Gamma_{\text{PO}}^{1/2} c$ ,  
 468 so the objective is  $w^\top N^{-1} w$ . By Cauchy–Schwarz,  $(w^\top w)^2 = (w^\top N^{-1/2} N^{1/2} w)^2 \leq$   
 469  $(w^\top N^{-1} w)(w^\top N w)$  and  $w^\top N w \leq \|w\|^2 \text{tr} N \leq \|w\|^2 B$ , hence  $w^\top N^{-1} w \geq \|w\|^2 / B =$   
 470  $c^\top \Gamma_{\text{PO}} c / B$ , attained at  $N^* = B w w^\top / \|w\|^2$ . Back-substituting with  $\Gamma_{\text{PO}}^{-1/2} w = c$  gives  
 471  $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$ : the optimal probe direction is  $c$  itself, the curvature pricing only its  
 472 scale. Verified: V4.1–V4.3 (formulas achieved; never beaten by 4000 random feasible designs  
 473 each).  $\square$

474 *Proof of Corollary C5.1 (price of isotropic jitter).* Isotropic  $M = (B / \text{tr} \Gamma_{\text{PO}}) I$  has  $\text{tr}(M^{-1}) =$   
 475  $d \text{tr}(\Gamma_{\text{PO}}) / B$ ; dividing by the optimum  $(\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$  gives  $F = d \text{tr}(\Gamma_{\text{PO}}) / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2$ , and  $F \geq 1$   
 476 with equality iff all eigenvalues are equal, by Cauchy–Schwarz applied to  $(\sum_i \sqrt{g_i} \cdot 1)^2 \leq d \sum_i g_i$ .  
 477 Verified: V4.4.  $\square$

478 *Proof of Lemma L3 (temporal shaping).* Under A3 the conditional log-likelihood of the responses  
 479 given the deployment path is  $-\sum_t (\tau_t - a - \theta^\top d_t)^2 / (2\sigma^2)$  up to constants; its Hessian in  $\theta$  is  
 480  $-\sum_t d_t d_t^\top / \sigma^2$ , a function of the empirical second moment only, whatever the temporal dependence  
 481 of  $\{d_t\}$ . Verified: V4.5.  $\square$

482 *Proof of Theorem T6 (Chebyshev unidentifiability).* The sensitivity of the family  $\tau(h) = C_0 +$   
 483  $C_1 e^{-ch}$  is  $s(h) = (1, e^{-ch}, -C_1 h e^{-ch})^\top$ . The functions  $\{1, e^{-ch}, h e^{-ch}\}$  form an extended  
 484 Chebyshev system on any interval (their Wronskian is  $c^3 e^{-2ch} \neq 0$  after row reduction; equivalently,  
 485 a nontrivial linear combination  $a + (b + \gamma h) e^{-ch}$  has at most two zeros counting multiplicity). For  
 486 any design  $\xi$  with  $k$  support points, the Fisher matrix  $M(\xi) = \sum_{i \leq k} w_i s(h_i) s(h_i)^\top$  has rank at  
 487 most  $k$ ; with  $k \leq 2 < 3$  it is singular, and that null space always loads on the  $c$ -sensitivity: writing  $n$   
 488 for the null vector of the two-point design matrix,  $n_3 = 0$  would force  $n_1 + n_2 e^{-ch_i} = 0$  at both  
 489 support points, hence  $n = 0$  for  $h_1 \neq h_2$ , a contradiction. By Corollary C1.2 the trajectory design

490 has at most two effective support points, which proves the claim; the existence of exactly-three-point  
 491 D-optimal designs on a trust-region interval is classical for Chebyshev systems (Karlin–Studden).  
 492 Verified: V5.1 (two-point determinants at  $10^{-17}$ ; three-point at  $6 \times 10^{-3}$ ).  $\square$

## 493 F. Safety under certainty equivalence

494 *Proof of Lemma L4 (perturbed modulus).* The corrected map is  $\Psi(h) = h + \eta [G(h) - \beta(h -$   
 495  $\psi) \hat{e}(h)]$  and the ideal map replaces  $\hat{e}$  by  $\varepsilon$ . With  $e = \hat{e} - \varepsilon$ ,  $\Psi'(h) - \Psi'_{\text{ideal}}(h) = -\eta \beta \frac{d}{dh} [(h -$   
 496  $\psi) e(h)] = -\eta \beta [e(h) + (h - \psi)e'(h)]$ , and taking suprema over the operating interval gives the  
 497 bound with  $|h - \psi|_{\max}$ ; reading the linearized modulus at the operating point  $h^*$  (where the certificate  
 498 is evaluated) localizes the constant to the registered  $|h^* - \psi|$ . Verified: V6.1 (nine-cell error grid; the  
 499 bound holds with maximal excess  $2.4 \times 10^{-11}$ ).  $\square$

500 *Proof of Remark R2 (open-loop neutrality).* For  $d_{t+1} = -m d_t + u_t$  with a history-independent  
 501 schedule  $\{u_t\}$ , two solutions with different initial conditions differ by the homogeneous solution  
 502  $(-m)^t(d_0 - d'_0)$ , which contracts at rate  $m$  regardless of the schedule (superposition). Verified: V6.2  
 503 (difference exactly 0 after 200 steps to machine precision).  $\square$

504 *Proof of Proposition P7.1 (safety under pessimism).* Let  $E$  be the event that the confidence sequence  
 505 contains the true response parameters simultaneously for all  $t$ ; by anytime-validity  $\mathbb{P}(E) \geq 1 - \delta$ .  
 506 On  $E$ , the true parameters lie in  $\Theta_t$  at every step, so the realized modulus is dominated by the rule's  
 507 worst case:  $|\rho_t| \leq \sup_{\theta \in \Theta_t} |\hat{\rho}(\theta)| \leq 1 - c_{\text{margin}}$  whenever correction is active, and when the rule  
 508 freezes, the blind loop's modulus  $m < 1 - c_{\text{margin}}$  (choose the margin below  $1 - m$ ) applies. No  
 509 union bound over time is needed: the inclusion is pathwise on  $E$ . (Conditional result: the hypothesis  
 510 is the confidence sequence's validity; constructing it from the D2 information accounting is routine,  
 511 and the necessity of the hypothesis is demonstrated numerically – an invalid, too-narrow sequence  
 512 produces violations.)  $\square$

## 513 G. The anchoring crossover

514 *Proof of Theorem T7 (horizon-dependent crossover).* Bias. For the symmetric secant with half-width  
 515  $w$ , fifth-order Taylor expansion of  $\tau(h^* \pm w)$  gives

$$\frac{\tau(h^* + w) - \tau(h^* - w)}{2w} = \tau'(h^*) + \frac{\tau'''(h^*)}{6} w^2 + O(w^4),$$

516 all even-order terms cancelling (symmetric probing is bias-optimal at its order; the second derivative  
 517 never enters). Verified: V7.1.

518 *Variance.* With  $n$  probe pairs ( $T = 2n$  observations) the slope estimator  $(\bar{y}_+ - \bar{y}_-)/(2w)$  has variance  
 519  $\frac{2\sigma^2/n}{4w^2} = \frac{\sigma^2}{2nw^2} = \frac{\sigma^2}{S}$  with  $S = \sum_t d_t^2 = 2nw^2 = Tw^2$ : variance depends on the design only  
 520 through its energy.

521 *Constrained optimum.* The value budget fixes  $S = 2B/\gamma_{\text{PO}}$ ; the horizon caps the number of  
 522 observations,  $Tw^2 \geq S$ , i.e.  $w^2 \geq 2B/(\gamma_{\text{PO}}T)$ ; bias is increasing in  $w$ , so the constrained optimum  
 523 sits at  $w^2 = 2B/(\gamma_{\text{PO}}T)$ , giving

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left( \frac{\tau'''(h^*)}{6} \right)^2 \left( \frac{2B}{\gamma_{\text{PO}}T} \right)^2.$$

524 Verified: V7.2–V7.4 (three  $(B, T)$  cells, within 2%).

525 *Crossover.* The anchored estimator has  $\text{MSE} = \kappa_p \gamma_{\text{PO}} \sigma^2 / (2B) + \delta_{\text{mis}}^2$ ; comparing and keeping the  
 526 dominant term gives the rule  $\delta_{\text{mis}} < |\tau'''(h^*)| B / (3\gamma_{\text{PO}}T)$  to leading order, with the exact two-term  
 527 form in the display above. The direction of the comparison on both sides of the threshold is verified:  
 528 V7.5, V7.6.  $\square$

## 529 H. ROI and instantiation

530 *Proof of Theorem T8 (ROI of self-knowledge).* With  $A = \frac{1}{2} \gamma_{\text{PO}} (h_{\text{SP}} - h_{\text{PO}})^2$ ,  $a = \frac{1}{2} \gamma_{\text{PO}} \kappa^2$ ,  $b =$   
 531  $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$ , the net present value of exploring to accuracy  $v$  is  $\text{NPV}(v) = (A - av)/\rho_{\text{disc}} - b/v$ , strictly

532 concave on  $(0, \infty)$  with  $\text{NPV}'(v) = -a/\rho_{\text{disc}} + b/v^2$ , giving  $v^* = \sqrt{b\rho_{\text{disc}}/a} = (\sigma/\kappa)\sqrt{\rho_{\text{disc}}}$   
 533 (the curvature cancels between  $a$  and  $b$ ). Substituting,  $\text{NPV}(v^*) = A/\rho_{\text{disc}} - 2\sqrt{ab/\rho_{\text{disc}}}$ , which  
 534 is positive iff  $A > 2\sqrt{ab\rho_{\text{disc}}}$ , i.e. iff  $\rho_{\text{disc}} < A^2/(4ab) = (h_{\text{SP}} - h_{\text{PO}})^4/(4\kappa^2\sigma^2)$ :  $\gamma_{\text{PO}}$  cancels  
 535 completely. Verified: V8.1, V8.2 (the break-even root to  $10^{-3}$  relative).  $\square$

536 *Proof of Proposition P9.1 (frontier consistency).* For any deterministic design,  $\text{Var}(\hat{\varepsilon} \mid \text{design}) \cdot$   
 537  $\frac{1}{2}\gamma_{\text{PO}}S = (\sigma^2/S) \cdot \frac{1}{2}\gamma_{\text{PO}}S = \frac{1}{2}\gamma_{\text{PO}}\sigma^2$ : the design energy cancels identically, whatever the schedule  
 538 – including the  $d_t \propto t^{-1/2}$  schedules with  $S_T = \Theta(\log T)$  of the Lai–Robbins adaptive-design line.  
 539 Verified: V8.3 (exact).  $\square$

540 *Proof of Proposition P9.2 (isotropy contrast).* Immediate from Corollary C5.1:  $F = 1$  iff the cur-  
 541 vature spectrum is flat – the isotropic (LQR-like) case in which naive exploration is optimal; any  
 542 curvature dispersion makes  $F > 1$  by the strict case of Cauchy–Schwarz. Cross-verified with  
 543 V4.4.  $\square$

544 *Proof of Theorem T9 (structure-proofness within reach).* Write  $x = h - h^*$ ,  $d(h) = x$ , and  $V =$   
 545  $\text{span}\{1, e^{-ch}, he^{-ch}\}$ , the span of the components of  $s$  (a bijective correspondence  $u \leftrightarrow \phi_u = u^\top s$   
 546 between  $\mathbb{R}^3$  and  $V$ , since  $C_1 \neq 0$ ). Differentiating  $s$  gives  $s'(h) = (0, -ce^{-ch}, C_1e^{-ch}(ch - 1))^\top$ ,  
 547 so  $g = -s'(h^*)$  componentwise, which is the stated estimand gradient.

548 *Step 1: one-dimensional reduction.* For  $M(\mu) \succ 0$  the generalized Rayleigh identity gives

$$g^\top M(\mu)^{-1}g = \sup_{u \neq 0} \frac{(u^\top g)^2}{u^\top M(\mu)u} = \sup_{\phi \in V} \frac{\phi'(h^*)^2}{\int \phi^2 d\mu} = \left( \inf \left\{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \right\} \right)^{-1},$$

549 using  $u^\top Mu = \int \phi_u^2 d\mu$  and  $u^\top g = -\phi'_u(h^*)$  (the sign is lost in the square). Hence

$$R(\mu) = \frac{\int d^2 d\mu}{\inf \left\{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \right\}}.$$

550 *Step 2: the candidate and its pointwise domination.* Let

$$\phi_0(h) = \frac{2}{c} - e^{-cx} \left( \frac{2}{c} + x \right) = \frac{2}{c} \cdot 1 - e^{ch^*} \left( \frac{2}{c} - h^* \right) e^{-ch} - e^{ch^*} (he^{-ch}) \in V,$$

551 the unique element of  $V$  with 2-jet  $(0, 1, 0)$  at  $h^*$  (the 2-jet map is invertible on  $V$ : in the basis  
 552  $\{1, e^{-ch}, he^{-ch}\}$  its determinant is  $c^2 e^{-2ch^*} \neq 0$ ); the expansion is  $\phi_0(x) = x - (c^2/6)x^3 + O(x^4)$ .  
 553 Writing  $f(x) = \frac{2}{c} - e^{-cx}(\frac{2}{c} + x)$  we claim

$$|f(x)| \leq |x| \quad \text{for all } x \geq -2/c, \quad \text{with equality iff } x \in \{0, -2/c\},$$

554 and  $|f(x)| > |x|$  strictly for every  $x < -2/c$ . Indeed: (a)  $v(x) = f(x) - x$  has  $v(0) = 0$  and  
 555  $v'(x) = e^{-cx}(1 + cx) - 1 < 0$  for  $x \neq 0$  (since  $1 + t < e^t$  for  $t = cx \neq 0$ );  $v' \leq 0$  with equality  
 556 only at the isolated point 0 makes  $v$  strictly decreasing on  $\mathbb{R}$ , so  $f > x$  on  $x < 0$  and  $f < x$  on  
 557  $x > 0$ . (b)  $f(0) = 0$  and  $f'(x) = e^{-cx}(1 + cx) > 0$  for  $x \geq 0$ , so  $f \geq 0$  on  $[0, \infty)$  and, with (a),  
 558  $0 \leq f \leq x$  there. (c)  $q(x) = f(x) + x$  has  $q(-2/c) = 0 = q(0)$  and  $q'(x) = e^{-cx}(1 + cx) + 1$ ,  
 559 where  $r(x) = e^{-cx}(1 + cx)$  is strictly increasing on  $x < 0$  ( $r'(x) = -c^2 x e^{-cx} > 0$ ) from  
 560  $r(-2/c) = -e^2 < -1$  to  $r(0) = 1$ ; so  $q'$  changes sign exactly once on  $(-2/c, 0)$ , from negative  
 561 to positive, and  $q \leq 0$  there with equality only at the endpoints; with (a) ( $f \geq x$ , equality only  
 562 at 0) this gives  $x \leq f \leq -x$  on  $[-2/c, 0]$ . (d) For  $x < -2/c$ ,  $r(x) < -e^2$  gives  $q' < 0$ , so  
 563 moving left from  $-2/c$  the function  $q$  increases above 0:  $f(x) > -x = |x| > 0$ , and in fact  
 564  $f(x) = 2/c + e^{c|x|}(|x| - 2/c)$  grows exponentially.

565 *Step 3: the bound.* If  $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$ , Step 2 gives  $\int \phi_0^2 d\mu \leq \int d^2 d\mu$ , so by Step 1,  
 566  $R(\mu) \geq \int d^2 d\mu / \int \phi_0^2 d\mu \geq 1$ . Strictness: equality in Step 2 holds only at  $x \in \{0, -2/c\}$ , so any  
 567  $\mu$  placing positive mass off  $\{h^*, h^* - 2/c\}$  (in particular any with three distinct support points)  
 568 has  $\int \phi_0^2 d\mu < \int d^2 d\mu$  and  $R(\mu) > 1$ . If  $M(\mu)$  is singular, two cases:  $g \notin \text{range } M(\mu)$  makes  
 569  $\varepsilon(h^*)$  non-estimable (infinite bound);  $g \in \text{range } M(\mu)$ , which does occur for special two-point  
 570 designs (a one-parameter family of in-reach pairs; their finite pseudo-inverse products were checked  
 571 to obey the bound, e.g.  $R = 1.063$  at the pair  $\{1.627, 2.127\}$ ), keeps the sup-representation valid

over  $\{u : u^\top M u > 0\}$ , and  $\int \phi_0^2 d\mu = 0$  would put  $u_0 \in \text{null } M$  with  $u_0^\top g = -1 \neq 0$ , contradicting  $g \in \text{range } M$ ; so  $\int \phi_0^2 d\mu > 0$  and the bound goes through verbatim. Throughout, the Cramér–Rao functional bounds unbiased estimation, and all regular estimators in the local-asymptotic-minimax sense.

*Step 4: the infimum is 1.* The Cramér–Rao functional is invariant under  $C^1$  reparametrization, and the 2-jet map  $\theta \mapsto (\tau_\theta(h^*), \varepsilon(h^*), \tau_\theta''(h^*))$  is a local diffeomorphism: its Jacobian  $[s(h^*) - s'(h^*) \ s''(h^*)]$  has determinant  $C_1 c^2 e^{-2ch^*} \neq 0$  (direct computation; V9.4). In jet coordinates  $\eta$  the model reads  $\tau = \eta_1 - \eta_2 x + \frac{1}{2} \eta_3 x^2 + O(x^3)$  with sensitivity  $\tilde{s}(x) = (1, -x, x^2/2)^\top + O(x^3)$ , and the estimand is  $\eta_2$ . For the symmetric three-point design  $\mu_\delta$  uniform on  $\{h^* - \delta, h^*, h^* + \delta\}$ , scale by  $D = \text{diag}(1, \delta, \delta^2)$ :  $M_\eta(\mu_\delta) = D \hat{H}_\delta D$  with  $\hat{H}_\delta \rightarrow \hat{H}_0$  invertible (the exact quadratic-model moment matrix), so  $(M_\eta^{-1})_{22} = \delta^{-2} (\hat{H}_\delta^{-1})_{22}$  and  $R(\mu_\delta) \rightarrow R_{\text{quad}}$ . In the exact quadratic model, for *any* symmetric design (odd moments  $m_1 = m_3 = 0$ ), the  $(2, 2)$  cofactor gives  $(M_\eta^{-1})_{22} = 1/m_2$  and  $R_{\text{quad}} = m_2 \cdot (1/m_2) = 1$ . Hence  $R(\mu_\delta) \rightarrow 1$  (measured rate  $R(\mu_\delta) - 1 = O(\delta^2)$ ; V9.2), the infimum is 1, and by Step 3 it is not attained.

*Step 5: the reach is exact.* Below  $h^* - 2/c$  the domination of Step 2 fails strictly, and the failure is realized by a design, not only by the candidate: the frozen four-point witness (support  $\{1.8726, 1.8665, 1.3073, 0.0500\}$ , weights  $\{0.003329, 0.955702, 0.040697, 0.000272\}$  after normalization, in the register market  $c = 1.5$ , frozen anchor  $h^* = 1.8461$ ) has  $R = 0.851650 \dots$ , verified at 50-digit precision (mpmath) and reproduced in float64 by `run_open1.py` (V9.2). All its support except the probe at 0.05 lies within reach: one vanishing-weight below-reach probe breaks the floor. Symmetric pair-plus-far-probe families do *not* break it (their ratio approaches 1 from above as the probe weight vanishes); the violation requires the asymmetric cluster, which is why coarse searches missed it. Verified: V9.1–V9.4.  $\square$

*Proof of Proposition P9.3 (separable multi-bond curvature).* In the separable case the objective is a sum of per-bond objectives  $\Phi_a(h_a)$ , each of the scalar form analyzed in the single-book theory of Nagarajan and Ashok [2026], so the Hessian is diagonal and the scalar computation applies per bond: differentiating  $\Phi_a(h) = P[h A_a e^{-k_a h} \rho + h \tau_a(h) - \psi_a \tau_a(h) - w_a(h - h_{\text{ref}})^2]$  twice and collecting terms with  $\tau_a' = -\varepsilon_a$ ,  $\tau_a'' = c_t \varepsilon_a$  yields  $-\Phi_a'' = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a)$ , which is the claimed diagonal entry. In the factor-coupled case the off-diagonal response contributes additional terms of first order in  $\|E_{\text{offdiag}}\|$ , stated as a construction with its error term (not claimed exact). Verified: V8.4 (finite-difference Hessians of three random bonds, relative error below  $10^{-4}$ ).  $\square$

## 603 C Complete derivations

This appendix reproduces the full derivation record behind every register result: the model’s closed forms, both budgets, and every intermediate computation, including the measurement-forced corrections recorded where they happened. Sections carry the derivation-record identifiers (D0–D9) that the register and Appendices A–B cite.

### 608 Notation, assumptions, and the two anchoring conventions

This subsection is the canonical register for the derivation record D0 through D9, with D3b the adaptive companion of D3: D0 is the model and cost lemma, D1 information saturation, D2 the exchange rate, D3 the minimax lower bounds, D4 and D5 the design geometry, D6 safe certainty equivalence, D7 the anchoring crossover, and D8 and D9 the ROI analysis and the instantiation. Every symbol is defined once, here; every assumption is numbered here and is cited by number in the derivation sections and in the register statements (T1, L2,  $\dots$ ). Where a later section restates a condition, this register is authoritative. The formal proofs of the register statements are collected in the proofs appendix.

**Notation.** The symbol table is reproduced in three blocks: the retraining loop and its objective, the stochastic and informational quantities, and the symbols specific to the design and economic analyses. The third column records where each symbol is introduced: a derivation section (D0 through D9), the base structural market theory (cited as REFLEX [Nagarajan and Ashok, 2026] with its section numbers, a third-party source), or an entry of the assumption register below (the trust-region radius

622  $r$  is fixed by A4). Two register conventions carry over unchanged: all logarithms are natural, and  
623 a prime denotes the derivative of a scalar function (in the ASCII sources it also denotes the matrix  
624 transpose; in this appendix transposes are written  $(\cdot)^\top$ , so the prime is unambiguous).

| Symbol                                 | Meaning                                                     | Defined in / source                               |
|----------------------------------------|-------------------------------------------------------------|---------------------------------------------------|
| $h, h_t$                               | deployed decision (half-spread); its value at step $t$      | D0                                                |
| $h^*$                                  | performatively stable point (RRM fixed point)               | REFLEX [Nagarajan and Ashok, 2026]                |
| $h_{SP}, h_{PO}$                       | stable point ( $h_{SP} = h^*$ ); performative optimum       | REFLEX [Nagarajan and Ashok, 2026]                |
| $d_t$                                  | deviation $d_t = h_t - h^*$                                 | D0                                                |
| $m$                                    | retraining modulus $m = \varepsilon\beta/\gamma \in [0, 1)$ | REFLEX [Nagarajan and Ashok, 2026]                |
| 625 $\gamma, \beta, \varepsilon, \psi$ | curvature, smoothness, response slope, adverse severity     | REFLEX [Nagarajan and Ashok, 2026] (closed forms) |
| $\Phi$                                 | true performative objective $\Phi(h) = J(h; \tau(h))$       | REFLEX [Nagarajan and Ashok, 2026]                |
| $\delta_0$                             | $\Phi'(h^*) = -\beta(h^* - \psi)\varepsilon(h^*)$ , nonzero | D0 Sec. 1                                         |
| $\gamma_{PO}$                          | $-\Phi''$ at the operating point                            | REFLEX [Nagarajan and Ashok, 2026].1              |
| $M$                                    | $d$ -dimensional loop Jacobian, $\rho(M) < 1$               | D1 Sec. 2                                         |
| $P, P_u$                               | discrete-Lyapunov energy matrices                           | D1 Sec. 2                                         |
| $\Gamma_{PO}$                          | $d$ -dimensional objective curvature matrix                 | D9                                                |

| Symbol           | Meaning                                                               | Defined in / source |
|------------------|-----------------------------------------------------------------------|---------------------|
| $\sigma$         | response-noise standard deviation ( $\sigma_\tau$ in REFLEX)          | D0                  |
| $\sigma_e, u_t$  | exploration intensity; exploration input                              | D0                  |
| $\xi_t, \zeta_t$ | deployment and response noise (iid, mean zero)                        | D0                  |
| $\phi$           | noise-feedback gain (observed noise flowing into the next deployment) | D2 Sec. 4           |
| $v$              | stationary deviation variance $\sigma_e^2/(1 - m^2)$                  | D2                  |
| $v_\phi$         | the same with feedback: $(\sigma_e^2 + \phi^2\sigma^2)/(1 - m^2)$     | D2 Sec. 4           |
| 626 $S_{xx}$     | centered design energy $\sum_t (d_t - \bar{d})^2$                     | D2                  |
| $D_T$            | deviation budget $\sum_t d_t^2$                                       | D0 Sec. 2           |
| $C_T$            | value budget: expected incremental cost (L1)                          | D0 Secs. 2–3        |
| $I_T$            | Fisher information for $\varepsilon$                                  | D1/D3               |
| $\mathcal{F}_t$  | history sigma-field $\sigma(d_0, u_{<t}, \tau_{<t})$                  | D0                  |
| $\delta, s$      | two-point prior half-width; its sign (minimax)                        | D3                  |
| $g_1$            | $\beta h^* - \psi $ , coefficient of the unknown linear part          | D3                  |
| TV, KL           | total variation; Kullback–Leibler divergence                          | D3                  |
| $r$              | trust-region radius on $ d_t $                                        | A4                  |

| Symbol                | Meaning                                                                                                                             | Defined in / source |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| $M$ (design)          | exploration second moment $\mathbb{E}[dd^\top]$ ; appears only in D4                                                                | D4                  |
| $B$                   | value budget in the design problems                                                                                                 | D4                  |
| $F$                   | curvature dispersion                                                                                                                | D4 (C5.1)           |
| $c$                   | $d \operatorname{tr}(\Gamma_{\text{PO}})/(\operatorname{tr} \Gamma_{\text{PO}}^{1/2})^2$ correction-direction vector (c-optimality) | D4 Sec. 3           |
| $s(h)$                | structural-family sensitivity vector, $p = 3$                                                                                       | D5                  |
| $w$                   | probe half-width (crossover analysis)                                                                                               | D7                  |
| $\delta_{\text{mis}}$ | $C^1$ misspecification of the structural family                                                                                     | D6/D7               |
| $\kappa$              | correction sensitivity $ dh_{\text{PO}}/d\varepsilon $                                                                              | D8                  |
| $\rho_{\text{disc}}$  | discount rate (ROI analysis)                                                                                                        | D8                  |

Three deliberate symbol reuses deserve a caveat, recorded here so that no derivation needs to repeat it. First,  $M$  names both the loop Jacobian and the exploration second moment; the design  $M$  occurs only in D4/D5, and context disambiguates the two. Second,  $\kappa$  is the correction sensitivity  $|dh_{\text{PO}}/d\varepsilon|$  of D8/D9, not a multi-dealer spillover parameter: no such symbol occurs in this paper. Third,  $\phi$  is the noise-feedback gain of D2, while the T9 material in D3 separately writes  $\phi_u$  and  $\phi_0$  for elements of the sensitivity span, a self-contained reuse local to that subsection.

**Assumption register.** Assumptions are cited by number (A1 through A6) throughout the record.

- **A1 (local quadratic scope).**  $\Phi$  is  $C^3$  and strongly concave on the operating interval; all expansions are taken at  $h^*$ , with the third-order remainder controlled through  $L_3 = \sup|\Phi'''|$ . The identities of the record are exact for the quadratic model and first-order accurate otherwise. As a matter of method, the nonlinear deviation is measured (the drift study), never assumed away.
- **A2 (exploration classes).** All policies are non-anticipating:  $u_t$  is  $\mathcal{F}_t$ -measurable. Two subclasses are used. *A2-sym*, conditionally symmetric exploration,  $\mathbb{E}[u_t | \mathcal{F}_t] = 0$ , under which the linear term of L1 vanishes in expectation. *A2-adaptive*, unrestricted non-anticipating exploration, under which costs are measured from the certainty-equivalent anchor (D0 Sec. 4; convention 2 below) and the exploitation lemma L2 of D3b applies.
- **A3 (noise exogeneity, with its named relaxation).** The response noise  $\zeta_t$  is iid, mean zero, with variance  $\sigma^2$ , independent of  $\mathcal{F}_t$ . Relaxation A3': the feedback channel  $\phi \zeta_t$  enters  $d_{t+1}$ . Under A3' the exact  $O(1/T)$  bias of D2 Sec. 4 applies, with its  $(3m - 1)$  sign structure (P2.2), and “exploration-dominant design” means  $\sigma_e^2 \gg \phi^2 \sigma^2$ .
- **A4 (trust region).**  $|d_t| \leq r$  pathwise. The radius  $r$  enters the constants of D3b (L2, T4), the feasible set of the design problems (T5), and the basin argument of D6.
- **A5 (stationary-scale exploration; D3b only).**  $S_T = \Theta(T)$ , with  $S_T = \sum_{t \leq T} d_t^2$  the cumulative design energy. The transient-dominated regime  $S_T = o(T)$  is governed by D1 (T1) instead.
- **A6 (stable base loop).**  $m < 1$  in the scalar case and  $\rho(M) < 1$  in the  $d$ -dimensional case, for all stationary-regime statements. Boundary behaviour ( $m = 1$ ) appears only in the degeneracy statement C1.2.

**The two anchoring conventions (audit-critical).**  $C_T$  is an incremental cost, and its value depends on the baseline against which the increment is measured. The record fixes one anchor per exploration class; these are the two audit-critical definitions.

*Convention 1 (incremental anchor; all A2-sym results).*  $C_T$  is measured relative to the jitter-free loop at  $h^*$ , that is, the same retraining loop run without exploration. Anchoring at  $h_{\text{PO}}$  instead adds the sunk echo-chamber term and provably destroys the invariance of D2 (T2): with anchor gap  $g = h^* - h_{\text{PO}}$ , the anchored exchange-rate product is inflated by exactly the factor

$$1 + \frac{g^2}{v},$$

664 a computation verified in D0. The formal proof is recorded in the proofs appendix.

665 *Convention 2 (certainty-equivalent anchor; all A2-adaptive results).*  $C_T$  is measured relative to the  
 666 deployment that is optimal under current posterior knowledge. Exploitation of learned structure is  
 667 thereby re-anchored away, and D3b (L2, T4) prices only the unlearned residual.

668 The two anchors coincide under A2-sym with an uninformative start, so the conventions agree on the  
 669 class where both are defined.

## 670 **D0: The model, the two budgets, and the cost-equivalence lemma (L1)**

671 This subsection records the local model, the two exploration budgets, and the computation behind the  
 672 cost-equivalence lemma L1. It is derived in full and verified numerically (`verify_all.py`, section  
 673 V0). Everything downstream (D1 through D8) consumes the definitions and the lemma established  
 674 here. Notation follows the REFLEX market model [Nagarajan and Ashok, 2026] and 1.2 ( $\gamma, \beta, \varepsilon, \psi$ ,  
 675  $h_{SP}, h_{PO}$ ).

676 **The local model.** Work in a neighbourhood of the performatively stable point  $h^*$  (the RRM fixed  
 677 point, the REFLEX market model [Nagarajan and Ashok, 2026], section 4) and write deviations  
 678  $d_t := h_t - h^*$ . The linearized closed loop is

$$\begin{aligned} d_{t+1} &= -m d_t + u_t && \text{(retraining cobweb plus exploration),} \\ \tau_t &= \tau(h^*) - \varepsilon d_t + \zeta_t && \text{(observed flow).} \end{aligned}$$

679 Here  $m = \varepsilon\beta/\gamma \in [0, 1]$  is the retraining modulus (closed form in the REFLEX market model  
 680 [Nagarajan and Ashok, 2026]). The exploration input  $u_t$  is chosen by the operator: it may  
 681 be iid noise ( $u_t = \sigma_e \xi_t$ ), a deterministic schedule, or adaptive, that is  $\mathcal{F}_t$ -measurable with  
 682  $\mathcal{F}_t = \sigma(d_0, u_0, \dots, u_{t-1}, \tau_0, \dots, \tau_{t-1})$ . The response noise  $\zeta_t$  is iid, mean zero, with variance  
 683  $\sigma^2$ , independent of  $\mathcal{F}_t$ ; the feedback violation of this independence assumption is quantified in the  
 684 exchange-rate derivation D2, section 4 (the feedback-bias result P2.2). The estimand is the response  
 685 slope  $\varepsilon$ , scalar here; the  $d$ -dimensional version (D4, behind T5) and the structural-family version (D5,  
 686 behind T6) are treated in their own derivation subsections.

687 The true performative objective  $\Phi(h) = J(h; \tau(h))$  (REFLEX theory 1.2) is smooth and strongly  
 688 concave on the operating interval, with

$$\Phi'(h^*) = \Delta(h^*) = -\beta(h^* - \psi)\varepsilon(h^*) =: \delta_0, \quad \Phi''(h^*) = -\gamma_{PO}$$

689 (the REFLEX market model [Nagarajan and Ashok, 2026], section 4.1). The slope  $\delta_0$  is nonzero  
 690 precisely because  $h^* \neq h_{PO}$ .

691 **The two budgets.** Two notions of exploration budget are used throughout:

$$\begin{aligned} D_T &:= \sum_{t \leq T} d_t^2 && \text{(deviation budget: realized design energy),} \\ C_T &:= \mathbb{E} \left[ \sum_{t \leq T} (\Phi(h^*) - \Phi(h_t)) \right] && \text{(value budget: expected incremental cost).} \end{aligned}$$

692  $D_T$  is the tracking-error form, a desk's deviation limit: it is observable, its meaning is policy-  
 693 independent, and it is the correct constraint for the exact minimax theorem T3 (derivation D3).  $C_T$  is  
 694 the realized-P&L form. Lemma L1 relates the two.

695 **Lemma L1 (cost equivalence).** For any exploration rule that is non-anticipating and conditionally  
 696 symmetric, meaning  $\mathbb{E}[u_t \mid \mathcal{F}_t] = 0$  (for example iid exploration, or a scheduled sign-symmetric  
 697 jitter), in the regime where third-order terms are negligible,

$$C_T = \frac{1}{2} \gamma_{PO} \mathbb{E}[D_T] + R_T, \quad |R_T| \leq \frac{L_3}{6} \mathbb{E} \left[ \sum_{t \leq T} |d_t|^3 \right],$$

698 with  $L_3 = \sup |\Phi'''|$  on the operating interval. The linear term of the expansion contributes exactly  
 699 zero to the expectation, and contributes a fluctuation of standard deviation  $|\delta_0| \sqrt{\text{Var}(\sum_{t \leq T} d_t)}$  to  
 700 the realized cost.

701 **Proof.** Taylor-expand  $\Phi$  at  $h^*$ :

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2} \gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

702 then sum over  $t \leq T$  and take expectations. For the linear term:  $d_t$  is built from  $\{u_s\}_{s < t}$  through  
 703 the linear recursion, so the mean obeys  $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t] + \mathbb{E}[u_t]$ ; conditional symmetry gives  
 704  $\mathbb{E}[u_t] = \mathbb{E}[\mathbb{E}[u_t | \mathcal{F}_t]] = 0$ , hence  $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0]$ . Starting at the operating point ( $d_0 = 0$ ),  
 705 or in the stationary regime,  $\mathbb{E}[d_t] = 0$  for all  $t$ , so  $\mathbb{E}[\sum_{t \leq T} \delta_0 d_t] = 0$ . The quadratic term yields  
 706  $\frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[D_T]$ , and the remainder is bounded by the stated third-order term.  $\square$

707 **Fluctuation decomposition (exact, and a corrected earlier claim).** The realized cost's variance  
 708 splits into two terms with zero cross term: the cross-moment sums built from  $\mathbb{E}[d_t d_s^2]$  vanish by the  
 709 sign symmetry of the stationary law. Exactly,

$$\text{Var}(C_T^{\text{real}}) = \delta_0^2 \text{Var}\left(\sum_{t \leq T} d_t\right) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}\left(\sum_{t \leq T} d_t^2\right),$$

710 with the long-run scales,  $v$  denoting the stationary variance of  $d_t$ ,

$$\begin{aligned} \text{Var}\left(\sum_{t \leq T} d_t\right) &\sim T v \frac{1-m}{1+m} && \text{(long-run variance: negative autocorrelation helps),} \\ \text{Var}\left(\sum_{t \leq T} d_t^2\right) &\sim T \frac{2v^2(1+m^2)}{1-m^2} && \text{(Gaussian stationary AR(1)).} \end{aligned}$$

711 A caveat, recorded per the falsification convention: an earlier draft asserted that the linear term always  
 712 dominates. The first verification run falsified that at moderate  $\delta_0$  (measured sd ratio 1.62, exactly  
 713 reproduced by the two-term formula above), and the claim is now stated as the full decomposition.  
 714 Which term dominates is decided by comparing  $\delta_0^2(1-m)/(1+m)$  against  $(\gamma_{\text{PO}}^2 v/2)(1+m^2)/(1-m^2)$ ,  
 715 both computable. V0 now verifies the decomposition itself.

716 **Why the anchoring is load-bearing (recorded from the audit).** Anchoring the cost at the perfor-  
 717 mative optimum  $h_{\text{PO}}$  instead of  $h^*$  adds terms of the type  $T(h^* - h_{\text{PO}})^2$  that are paid whether or not  
 718 the operator explores: the sunk echo-chamber cost of blindness (REFLEX theory 1.2, 6b). Numeri-  
 719 cally (verify\_m2\_identity.py, Claim 3): writing  $g = h^* - h_{\text{PO}}$  for the gap, the mis-anchored  
 720 product equals

$$\frac{1}{2} \gamma_{\text{PO}} \sigma^2 \left(1 + \frac{g^2}{v}\right),$$

721 which destroys the exchange-rate invariance of D2 (theorem T2). The incremental anchor at  $h^*$  is the  
 722 definition under which every later theorem is true.

723 **The certainty-equivalent anchor (needed for the adaptive minimax bound T4, derivation D3).**  
 724 For adaptive policies the symmetric-exploration condition can be violated deliberately: since  $\delta_0 \neq 0$ ,  
 725 drifting toward  $h_{\text{PO}}$  earns first-order value. Two observations discipline this. First, decompose  
 726  $\Phi'(h^*)$  as a function of the estimand:

$$\varphi(\varepsilon) := \Phi'(h^*; \varepsilon) = -\beta(h^* - \psi)\varepsilon,$$

727 which is linear in  $\varepsilon$ . Under a prior centred at  $\bar{\varepsilon}$ , split  $\varphi(\varepsilon) = \varphi(\bar{\varepsilon}) + \varphi'(\bar{\varepsilon})(\varepsilon - \bar{\varepsilon})$ . Second, the  
 728 first (known) part is exploitable without any learning: it is a pure consequence of already-known  
 729 information, that is, of choosing the anchor wrongly. Define therefore the certainty-equivalent anchor  
 730  $h_{\text{CE}}^*$ : the deployment that is optimal under current knowledge (the posterior mean), with exploration  
 731 cost measured relative to the trajectory that sits at  $h_{\text{CE}}^*$ . Relative to that anchor the residual linear  
 732 coefficient is

$$\varphi'(\bar{\varepsilon})(\varepsilon - \bar{\varepsilon}) = -\beta(h^* - \psi)(\varepsilon - \bar{\varepsilon}),$$

733 which has zero prior mean: only genuinely unknown structure can be exploited, and the exploitation-  
 734 information lemma L2 (in D3, feeding T4) bounds that exploitation by the information already  
 735 purchased. This definition also matches practice: a desk's exploration cost is measured against its  
 736 own current best policy, not against an oracle's.

737 **Numerical verification (V0).** Three checks close the record for this subsection.

- 738 •  $\mathbb{E}[C_T] = \frac{1}{2} \gamma_{PO} \mathbb{E}[D_T]$  holds within Monte Carlo error, for a quadratic-plus-linear objective  
739 with  $\delta_0 \neq 0$ , under iid exploration.
- 740 • The realized-cost fluctuation scale matches  $|\delta_0| \text{sd}(\sum_{t \leq T} d_t)$ : the linear term is a variance  
741 contribution, not a bias.
- 742 • A deliberately asymmetric (drifting) exploration rule breaks the lemma in the predicted di-  
743 rection and by the predicted first-order amount  $-\delta_0 \mathbb{E}[\sum_{t \leq T} d_t]$ : the conditional-symmetry  
744 condition of L1 is necessary, not decorative.

745 **D1: Information saturation by the Lyapunov identity (T1, C1.1, C1.2, R1)**

746 This subsection records the derivation behind T1 and its corollaries C1.1 and C1.2, together with  
747 the non-normal refinement R1. Status of the record: the scalar closed form is verified to  $10^{-16}$ , and  
748 the  $d$ -dimensional identity, including the non-normal amplification, is verified numerically (V1); the  
749 proofs are complete.

750 **Scalar warm-up (the verified base case).** With no exploration ( $u_t = 0$ ) and modulus  $m < 1$ , the  
751 deviation obeys  $d_t = (-m)^t d_0$ , so the design energy accumulated by horizon  $T$  is

$$D_T = \sum_{t=0}^T d_t^2 = d_0^2 \frac{1 - m^{2(T+1)}}{1 - m^2} \rightarrow \frac{d_0^2}{1 - m^2} \quad (T \rightarrow \infty).$$

752 The Fisher information for the response slope  $\varepsilon$  from OLS on the trajectory is  $S_{xx}/\sigma^2 \leq D_\infty/\sigma^2$ :  
753 bounded uniformly in the horizon. Hence, by Cramér–Rao, for any unbiased estimator and at every  
754 horizon,

$$\text{Var}(\hat{\varepsilon}) \geq \frac{\sigma^2 (1 - m^2)}{d_0^2}.$$

755 This is the scalar case of T1; with  $d_0 = h_0 - h^*$  the cap  $d_0^2/(1 - m^2)$  is the quantity in the register  
756 statement of C1.1.

757 **C1.1 (safety implies blindness).** The cap  $d_0^2/(1 - m^2)$  is strictly increasing in  $m$ : the fast-  
758 contracting (safe) loop reveals least about its own performativity, and identifiability is a near-instability  
759 phenomenon. This is immediate from the closed form above; the economic content is stated in the  
760 paper body.

761 **C1.2 (design degeneracy).** At  $m = 1$  the noiseless iterates form the period-two orbit  $\{+d_0, -d_0\}$ ;  
762 for  $m < 1$  the support collapses geometrically onto  $\{0\}$ . In either case the empirical design measure  
763 has at most two effective support points, which by the Chebyshev counting argument of D5 (theorem  
764 T6) cannot identify a three-parameter response family. Curvature is invisible to retraining trajectories  
765 at every  $m$ .

766 **The  $d$ -dimensional identity via the discrete Lyapunov equation.** Let the linearized joint loop be  
767  $d_{t+1} = M d_t$  with  $M \in \mathbb{R}^{d \times d}$  and  $\rho(M) < 1$ ; in the multi-bond and multi-dealer instantiations  $M$   
768 comes from the REFLEX market model [Nagarajan and Ashok, 2026] (the base structural market  
769 theory) and is generally not symmetric.

770 **Theorem (exact energy; T1).** The total design energy is the quadratic form

$$E(d_0) := \sum_{t \geq 0} \|M^t d_0\|^2 = d_0^\top P d_0,$$

771 where  $P$  is the unique positive semidefinite solution of the discrete Lyapunov equation

$$P = I + M^\top P M.$$

772 **Proof.** Set  $P := \sum_{t \geq 0} (M^\top)^t M^t$ ; the series converges absolutely if and only if  $\rho(M) < 1$  (Gelfand’s  
773 spectral radius formula). Then

$$M^\top P M = \sum_{t \geq 0} (M^\top)^{t+1} M^{t+1} = P - I,$$

774 so  $P$  solves the equation. Uniqueness: the difference  $Q$  of two solutions satisfies  $Q = M^\top Q M =$   
 775  $(M^\top)^t Q M^t \rightarrow 0$ . Finally,  $d_0^\top P d_0 = \sum_{t \geq 0} \|M^t d_0\|^2$  by expanding the sum term by term.  $\square$

776 Scalar check:  $P = 1/(1 - m^2)$ , recovering the warm-up above.

777 **Directional information.** For the response slope in a direction  $u$  (the observation  $\tau_t = \tau^* -$   
 778  $\varepsilon u^\top d_t + \zeta_t$  probes the component along  $u$ ), the relevant energy is

$$\sum_{t \geq 0} (u^\top M^t d_0)^2 = d_0^\top P_u d_0, \quad P_u = uu^\top + M^\top P_u M,$$

779 the same Lyapunov structure with a rank-one right-hand side. By Cramér–Rao this yields  $\text{Var}(\hat{\varepsilon}) \geq$   
 780  $\sigma^2 (d_0^\top P_u d_0)^{-1}$  for the component along any direction  $u$ , which is the bound stated in T1. Hence the  
 781 Fisher information is bounded in every direction, and the information operator satisfies  $\mathcal{I} \preceq P_{\text{full}}/\sigma^2$ ,  
 782 where  $P_{\text{full}}$  solves the matrix-valued Lyapunov equation. Saturation is not a scalar accident.

783 **Non-normality: transiently amplified information (R1).** If  $M$  is normal,  $P$  has eigenvalues  
 784  $1/(1 - \lambda_i^2)$  and the energy is governed by the spectrum alone. If  $M$  is non-normal (heterogeneous  
 785 multi-dealer Jacobians are), transient growth inflates the energy beyond the spectral prediction:

$$E(d_0) \leq \kappa(V)^2 \frac{\|d_0\|^2}{1 - \rho(M)^2},$$

786 where  $V$  is the eigenbasis of  $M$  and  $\kappa(V)$  its condition number, and the inequality is achieved in  
 787 order: for the Jordan-type family

$$M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix},$$

788 the energy along the first coordinate grows like  $a^2/(1 - m^2)^3$  for large  $a$ , verified numerically in V1  
 789 against the exact Lyapunov solve. Reading: badly-conditioned ecosystems leak extra information  
 790 about themselves during transients, the free-information remark of the paper, now with an exact  
 791 constant ( $P$  itself). The paper states the exact  $P$ -form and keeps the pseudospectral refinement for  
 792 the journal version.

793 **Scope: what T1 does not say.** Two caveats are part of the record and are stated in the paper.

- 794 • With exploration on, saturation is replaced by the exchange rate of D2 (T2): the transient  
 795 information  $d_0^\top P d_0/\sigma^2$  is the free part; everything after is bought.
- 796 • The bound is for the linearized loop; the simulator’s nonlinear regimes are handled empiri-  
 797 cally (the drift study, under the D2/V2 protocol).

798 **Numerical verification (V1).** Four checks anchor the derivation. (i) The scalar closed form against  
 799 the direct sum: exact to floating point. (ii) Random stable  $M$  of sizes  $2 \times 2$  and  $5 \times 5$ : the direct  
 800 energy sum against the Lyapunov solve (a `scipy`-free fixed-point iteration), agreement to  $10^{-12}$ . (iii)  
 801 Non-normal amplification: the energy for the family  $M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix}$  grows as  $a^2$ , far exceeding the  
 802 normal-matrix bound  $1/(1 - m^2)$ , and matches  $d_0^\top P d_0$  exactly. (iv) The directional version: the  
 803 rank-one Lyapunov solution  $P_u$  against the direct directional sum.

## 804 **D2: The exchange-rate identity and its concentration (T2, P2.1, P2.2)**

805 This subsection records the derivations behind T2 (the exchange-rate identity in pathwise form), P2.1  
 806 (its concentration), and P2.2 (the feedback-bias constant). Status of record: the identity was verified  
 807 to zero relative error by a pre-existing check; the pathwise refinement, the concentration rate, and the  
 808 feedback-bias constant are derived here and verified in V2. The formal statements live in the register  
 809 of theorem statements, and the proofs in the proofs appendix.

810 **Setup.** The stationary jittered loop is

$$d_{t+1} = -m d_t + \sigma_e \xi_t, \quad v = \text{Var}(d_t) = \frac{\sigma_e^2}{1 - m^2}.$$

811 OLS of the response  $\tau_t$  on the quote  $h_t$  has conditional-on-design variance  $\text{Var}(\hat{\varepsilon} \mid \text{design}) = \sigma^2/S_{xx}$   
 812 with  $S_{xx} = \sum_{t \leq T} (d_t - \bar{d})^2$ , and the cost of Lemma L1 is  $\frac{1}{2} \gamma_{\text{PO}} \sum_{t \leq T} d_t^2$ .

813 **The identity, upgraded from expectation to pathwise (T2). Proposition (pathwise identity).**  
814 On every realization,

$$\begin{aligned}\text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2} \gamma_{\text{PO}} \sum_{t \leq T} d_t^2 &= \frac{1}{2} \gamma_{\text{PO}} \sigma^2 \frac{\sum_{t \leq T} d_t^2}{S_{xx}} \\ &= \frac{1}{2} \gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T \bar{d}^2}{S_{xx}}\right).\end{aligned}$$

815 Since  $\bar{d} \rightarrow 0$  almost surely at rate  $T^{-1/2}$  in the stationary regime, the product equals  $\frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 +$   
816  $O_P(1/T))$  pathwise: a strictly stronger statement than the in-expectation identity, and the form the  
817 experiments actually measure. The correction term  $T \bar{d}^2 / S_{xx}$  is nonnegative, so the realized product  
818 can only sit above the constant, by a vanishing amount.

819 **Invariance.** Neither  $\sigma_e$ , nor  $m$ , nor  $T$  appears: more jitter buys information and burns value at the  
820 same rate; a larger modulus amplifies the jitter into more excitation and more cost at the same rate.  
821 The constant  $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$  is the exchange rate between self-knowledge and value.

822 **Concentration: the identity is observable (P2.1).** The realized value cost (Lemma L1) carries  
823 the linear-term fluctuation  $-\delta_0 \sum_{t \leq T} d_t$ . In the stationary regime  $\sum_t d_t$  is a mean-zero weakly  
824 dependent sum, so

$$\frac{\text{sd}(\text{realized } C_T)}{\mathbb{E}[C_T]} = O(T^{-1/2}),$$

825 with the explicit leading constant given by the AR(1) long-run variance  $v(1 + \rho)/(1 - \rho)$  at lag-one  
826 autocorrelation  $\rho = -m$ , that is  $v(1 - m)/(1 + m)$ :

$$\text{Var}\left(\sum_{t \leq T} d_t\right) = T v \frac{1 - m}{1 + m} (1 + o(1)).$$

827 The negative autocorrelation of the cobweb shrinks the fluctuation below the iid level  $T v$ : a small  
828 bonus worth one sentence in the paper. Hence the measured product concentrates around the invariant  
829 at rate  $T^{-1/2}$ . Verified in V2: the seed-variance of the measured product scales as  $1/T$  (it halves  
830 when  $T$  doubles, a factor-4 drop when  $T$  quadruples, within Monte-Carlo tolerance).

831 **Detail:  $S_{xx}$  versus  $\sum_t d_t^2$ .** The estimator uses the centered  $S_{xx}$ ; the cost uses the raw  $\sum_t d_t^2$ . Their  
832 ratio is

$$\frac{\sum_{t \leq T} d_t^2}{S_{xx}} = 1 + \frac{T \bar{d}^2}{S_{xx}} = 1 + O_P(1/T),$$

833 as above. Statements in the paper use whichever form is natural and record this  $O(1/T)$  equivalence  
834 once, in the present form.

835 **The feedback (predetermined-regressor) bias, with its constant (P2.2).** In the real loop the  
836 observed response feeds the next deployment. Model this as

$$d_{t+1} = -m d_t + \sigma_e \xi_t + \phi \zeta_t,$$

837 with  $\phi$  the noise-feedback gain: retraining on the realized flow moves the next quote by  $\phi \zeta_t$ . Now  $d_t$   
838 is correlated with past  $\zeta$ , so the regressor is predetermined but not strictly exogenous.

839 **Proposition (Stambaugh-type bias, full form, corrected by verification).** The OLS slope satisfies

$$\mathbb{E}[\hat{\varepsilon} - (-\varepsilon)] = \frac{\phi \sigma^2 (3m - 1)}{(1 - m^2) T v_\phi} (1 + o(1)), \quad v_\phi = \frac{\sigma_e^2 + \phi^2 \sigma^2}{1 - m^2},$$

840 where  $-\varepsilon$  is the true slope in this sign convention (written  $\varepsilon_{\text{true}}$  in P2.2) and  $v_\phi$  is the stationary  
841 variance of the feedback loop. The bias is  $O(1/T)$ , proportional to the feedback gain, and changes  
842 sign at  $m = 1/3$ : slow-contracting loops bias the slope estimate upward, fast-contracting ones  
843 downward, and at  $m = 1/3$  exactly the feedback bias vanishes at first order.

844 **Derivation, both terms.** (Honesty record, kept per the falsification convention: the first draft retained  
845 only the centering term and was falsified by V2; the covariance term below is the correction.) Write  
846  $\hat{b} - b = N/S$  with  $N = \sum_{t \leq T} (d_t - \bar{d}) \zeta_t$  and  $S = S_{xx}$ , and expand

$$\mathbb{E}\left[\frac{N}{S}\right] = \frac{\mathbb{E}[N]}{\mathbb{E}[S]} - \frac{\text{Cov}(N, S)}{\mathbb{E}[S]^2} + O(T^{-2}).$$

847 *Centering term.* Here  $\mathbb{E}[d_t \zeta_t] = 0$ , since  $\zeta_t$  is independent of the past-built  $d_t$ , so  $\mathbb{E}[N] =$   
848  $-\mathbb{E}[\bar{d} \sum_t \zeta_t]$ . With  $\mathbb{E}[d_s \zeta_t] = \phi \sigma^2 (-m)^{s-t-1}$  for  $s > t$ ,

$$\mathbb{E}[N] = -\frac{1}{T} \sum_{s>t} \phi \sigma^2 (-m)^{s-t-1} = -\frac{\phi \sigma^2}{1+m} + O(1/T).$$

849 *Covariance term (the one the first draft missed).* By the Gaussian fourth-moment (Wick) expansion,

$$\text{Cov}(N, S) = 2 \sum_{t,s} \mathbb{E}[\tilde{d}_t \tilde{d}_s] \mathbb{E}[\tilde{d}_s \zeta_t], \quad \tilde{d} = d - \bar{d};$$

850 substituting  $\mathbb{E}[\tilde{d}_t \tilde{d}_s] \simeq v_\phi (-m)^{|s-t|}$  and the cross-covariance above,

$$\text{Cov}(N, S) = 2 v_\phi \phi \sigma^2 T \sum_{k \geq 1} (-m)^k (-m)^{k-1} = -\frac{2 v_\phi \phi \sigma^2 T m}{1-m^2}.$$

851 *Combine.* With  $\mathbb{E}[S] \simeq T v_\phi$ ,

$$\text{bias} = \left[ -\frac{1}{1+m} + \frac{2m}{1-m^2} \right] \frac{\phi \sigma^2}{T v_\phi} = \frac{3m-1}{1-m^2} \cdot \frac{\phi \sigma^2}{T v_\phi},$$

852 which is P2.2. Verified in V2 at two moduli straddling the sign change:  $m = 0.5$  gives a positive bias  
853 and  $m = 0.2$  a negative one, each within Monte-Carlo tolerance of the constant. The sign change is  
854 the sharpest possible falsification test of the two-term structure, and it passes.

855 **Paper usage.** State the identity (T2) under exploration-dominant design; report the bias formula  
856 (P2.2) as the known, computable deviation, a checkable condition ( $\phi$  is estimable from the loop's  
857 own update rule), not an assumption.

858 **Verified numerically (V2).** The V2 checks for this subsection: (1) the pathwise product equals  
859  $\frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + T \bar{d}^2 / S_{xx})$  exactly; (2) product concentration: the seed-variance of the measured  
860 product scales as  $1/T$ , a factor-4 drop when  $T$  quadruples, within Monte-Carlo tolerance; (3) the  
861 long-run variance of  $\sum_t d_t$  matches  $T v (1-m)/(1+m)$ ; (4) the feedback bias: sign, magnitude,  
862 and  $1/T$  scaling of  $\phi \sigma^2 (3m-1)/((1-m^2) T v_\phi)$ .

### 863 **D3: Minimax lower bounds and the exact reach (T3, L2, T4, T9)**

864 This record derives the two lower bounds that turn the exchange-rate identity of D2 (register T2) from  
865 a property of one scheme into a property of the problem, and then the exact reach within which that  
866 floor is structure-proof against a policy that knows the response family (T9). T2 shows that a specific  
867 scheme, stationary symmetric jitter with ordinary least squares, achieves  $\text{Var}(\hat{\varepsilon}) \times \text{cost} = \frac{1}{2} \gamma_{\text{PO}} \sigma^2$ .  
868 The claim that makes this an exchange rate is that no adaptive exploration policy and no estimator  
869 does better. There are two versions of the bound, one per budget type: an exact deviation-budget  
870 bound (T3, derived in full and verified numerically in V3a) and a value-budget bound (T4), whose  
871 engine is the exploitation–information lemma (L2), proved below with its  $O(T^{-1/2})$  correction  
872 established and verified in V3b. Throughout, the formal proofs are recorded in the proofs appendix;  
873 this subsection records the derivations and the caveats attached to them.

874 **Deviation budget: van Trees over adaptive designs (T3).** Setting: the exploration policy is  
875 arbitrary and adaptive (each  $u_t$  any measurable function of the history), subject to the pathwise budget  
876  $\sum_{t \leq T} d_t^2 \leq D$  almost surely; the estimator  $\hat{\varepsilon}$  of  $\varepsilon$  is arbitrary, bias allowed. The claim (T3) is that  
877 the minimax risk over an interval of  $\varepsilon$  values of width  $w$  satisfies

$$\inf_{\text{policy, estimator}} \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\sigma^2}{D + \sigma^2(\pi/w)^2} = \frac{\sigma^2}{D} (1 - o(1)) \quad \text{as } D(w/\sigma)^2 \rightarrow \infty.$$

878 Multiplying by the budget cost  $\frac{1}{2}\gamma_{\text{PO}}D$  (cost equivalence, L1) gives

$$\sup_{\varepsilon} \text{Var}(\hat{\varepsilon}) \times \frac{1}{2}\gamma_{\text{PO}}D \geq \frac{1}{2}\gamma_{\text{PO}}\sigma^2(1 - o(1)) :$$

879 the exchange rate is a floor.

880 The derivation is the van Trees inequality (Bayesian Cramér–Rao, in the form of Gill and Levit, 1995)  
881 with a prior density  $\pi_0$  supported on the width- $w$  interval:

$$\mathbb{E}_{\pi_0} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{1}{\mathbb{E}_{\pi_0}[I_T(\varepsilon)] + I(\pi_0)}, \quad I(\pi_0) = \int \frac{(\pi'_0)^2}{\pi_0},$$

882 where  $I(\pi_0)$  is the prior information, equal to  $(\pi/w)^2$  for the cosine-squared prior (the minimizer of  
883 prior information on an interval of width  $w$ ), and  $I_T(\varepsilon)$  is the Fisher information of the adaptively  
884 designed experiment. The one nonstandard step is that  $I_T$  for an adaptive design is still bounded by  
885 the design energy: by the chain rule for Fisher information over the filtration,

$$I_T(\varepsilon) = \sum_{t \leq T} \mathbb{E}[I(\tau_t | \mathcal{F}_t; \varepsilon)] = \sum_{t \leq T} \frac{\mathbb{E}[d_t^2]}{\sigma^2},$$

886 because conditionally on  $\mathcal{F}_t$  the observation  $\tau_t$  is Gaussian with mean linear in  $\varepsilon$  (slope  $-d_t$ , with  $d_t$   
887  $\mathcal{F}_t$ -measurable): the policy can place information but cannot manufacture it. The pathwise budget  
888 then gives  $\mathbb{E}[I_T] \leq D/\sigma^2$ ; substituting, and taking the supremum over the interval ( $\sup \geq \mathbb{E}_{\pi_0}$ ),  
889 yields the bound.

890 Three remarks recorded with the derivation. (i) Biased estimators are covered: van Trees needs no  
891 unbiasedness. (ii) The bound is achieved, up to the  $o(1)$ , by two-point symmetric probing at the  
892 budget with least squares, tying back to the D2 identity (T2); this is the “attained with equality”  
893 clause of T3. (iii) The  $o(1)$  is explicit: it equals  $\sigma^2(\pi/w)^2/D$ .

894 **D3b, the adaptive companion. Value budget: the exploitation problem.** Under the value budget  
895 the policy pays  $\Phi(h_{\text{CE}}^*) - \Phi(h_t)$  per step (the certainty-equivalent anchor of D0). The residual linear  
896 coefficient at the anchor is

$$\phi'(\bar{\varepsilon})(\varepsilon - \bar{\varepsilon}) = -\beta(h^* - \psi)(\varepsilon - \bar{\varepsilon}),$$

897 so a policy could try to earn while exploring: guess the sign of  $\varepsilon - \bar{\varepsilon}$  and drift accordingly. The  
898 lemma below says this self-financing is bounded by the information already purchased: the policy  
899 cannot exploit what it has not yet paid to know.

900 Setting: symmetric two-point prior  $\varepsilon \in \{\bar{\varepsilon} - \delta, \bar{\varepsilon} + \delta\}$ ; write  $s = \text{sign}(\varepsilon - \bar{\varepsilon})$  and  $g_1 := \beta|h^* - \psi|$ ,  
901 the known coefficient of the unknown part. The policy is non-anticipating, with per-step trust region  
902  $|d_t| \leq r$ ; write  $S_t := \sum_{i \leq t} d_i^2$  for the design energy.

903 **The exploitation–information lemma (L2).** Two ingredients. First, the posterior-imbalance step:  
904 the policy’s knowledge of  $s$  at time  $t$  obeys

$$|\mathbb{E}[s | \mathcal{F}_t]| \leq \text{TV}(P_t^+, P_t^-) =: \text{TV}_t$$

905 (posterior imbalance is bounded by the distinguishability of the two environments; the starting prior  
906 is symmetric). Register L2 records the averaged form of this step as the exact identity  $\mathbb{E}|\mathbb{E}[s | \mathcal{F}_t]| =$   
907  $\text{TV}(P_t^+, P_t^-)$ . Second, the Pinsker step: the two environments differ only in the observation means,  
908 by  $2\delta d_i$  at step  $i$ , so

$$\text{KL}_t = \sum_{i \leq t} \frac{(2\delta d_i)^2}{2\sigma^2} = \frac{2\delta^2 S_t}{\sigma^2}, \quad \text{TV}_t \leq \sqrt{\text{KL}_t/2} = \frac{\delta\sqrt{S_t}}{\sigma}.$$

909 (For adaptive designs the same computation runs through the chain rule for KL over the filtration,  
910 with  $S_t$  replaced by  $\mathbb{E}S_t$ , exactly as in the Fisher computation for T3; this is the form registered in  
911 L2.)

912 Chaining these, the expected exploitation gain obeys, for any non-anticipating policy,

$$\begin{aligned}\mathbb{E}[\text{gain}_T] &:= \mathbb{E}\left[\sum_{t \leq T} g_1 \delta s d_t\right] \leq g_1 \delta \sum_{t \leq T} \mathbb{E}[\text{TV}_t | d_t|] \\ &\leq \frac{g_1 \delta^2}{\sigma} \sqrt{S_T} \sum_{t \leq T} \mathbb{E}|d_t| \leq \frac{g_1 \delta^2}{\sigma} S_T \sqrt{T},\end{aligned}$$

913 where the first inequality is the tower rule plus the posterior-imbalance step, the second uses  $\text{TV}_t \leq$   
 914  $\delta \sqrt{S_t}/\sigma \leq \delta \sqrt{S_T}/\sigma$ , and the last is Cauchy–Schwarz ( $\sum_{t \leq T} |d_t| \leq \sqrt{T} S_T$ ). For fully adaptive  
 915 designs the same chain, run with expected energies, gives the closed form recorded in L2,  $\mathbb{E}[\text{gain}_T] \leq$   
 916  $g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4} / \sigma^{1/2}$ ; both forms are  $O(\sigma \sqrt{T})$  at the minimax scale below. The formal proof  
 917 is recorded in the proofs appendix under L2.

918 **The minimax scale and the value-budget bound (T4).** The minimax-relevant prior scale is  
 919  $\delta = c \sigma / \sqrt{S_T}$ : this is the accuracy the budget can buy (any larger  $\delta$  is learned exactly, any smaller is  
 920 unlearnable). Substituting into the lemma,

$$\mathbb{E}[\text{gain}_T] \leq g_1 c^2 \sigma \sqrt{T}, \quad \text{while} \quad \mathbb{E}[C_T] = \frac{1}{2} \gamma_{\text{PO}} S_T \sim \frac{1}{2} \gamma_{\text{PO}} T v$$

921 (cost equivalence L1, stationary-scale exploration  $S_T = \Theta(T)$  with per-step exploration variance  $v$ ).  
 922 The exploitable fraction of the budget is therefore

$$\frac{\mathbb{E}[\text{gain}_T]}{\mathbb{E}[C_T]} \leq \frac{2 g_1 c^2 \sigma}{\gamma_{\text{PO}} v \sqrt{T}} = O(T^{-1/2}).$$

923 The Le Cam two-point reduction then quantifies the constant. At the minimax scale  $\text{TV}_T \leq c$ , so any  
 924 estimator’s worst-case risk over the pair obeys

$$\sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\delta^2}{2} (1 - \text{TV}_T) \geq \frac{\delta^2}{2} (1 - c).$$

925 Multiplying by  $\mathbb{E}[C_T] = \frac{1}{2} \gamma_{\text{PO}} S_T$ , using  $\delta^2 S_T = c^2 \sigma^2$ , and applying the exploitation rebate (the  
 926 value budget can be underpaid only by the exploited gain, an  $O(T^{-1/2})$  fraction by the display  
 927 above):

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{\gamma_{\text{PO}} \sigma^2}{4} c^2 (1 - c) (1 - O(T^{-1/2})).$$

928 The factor  $c^2(1 - c)$  is maximized at  $c = 2/3$ , with value  $4/27$ , giving the registered bound of T4:

$$\inf_{\text{policy, estimator}} \sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{\gamma_{\text{PO}} \sigma^2}{27} (1 - O(T^{-1/2})).$$

929 The sharp constant  $\frac{1}{27}$ , matching the achievable product of T2, that is  $\inf \sup \text{Var}(\hat{\varepsilon}) \times C_T \geq$   
 930  $\frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 - O(T^{-1/2}))$ , is conjectured and is part of OPEN-1: no simulated policy fell below it by  
 931 more than Monte-Carlo error (run\_open1.py, part B).

932 **Interpretation: the self-referential structure.** The profitable direction is  $s$ , the very thing being  
 933 estimated. Pinsker converts “how much the policy knows about  $s$ ” into “how much design energy it  
 934 has already spent”; any gain is therefore a rebate proportional to information already paid for, and the  
 935 rebate’s rate vanishes. Knowledge cannot be bootstrapped from its own purchase. To our knowledge  
 936 no analog of this lemma appears in the performative-prediction, pricing, or experimental-design  
 937 literatures (literature review items G1 to G3).

938 **Caveats recorded with the derivation.** (i) The constant in the  $O(T^{-1/2})$  is explicit,  
 939  $2 g_1 c^2 \sigma / (\gamma_{\text{PO}} v)$ , and blows up as  $v \rightarrow 0$ : a policy that barely explores can have its tiny cost  
 940 substantially rebated. This is consistent, since in that regime both sides of the product are domi-  
 941 nated by the transient (D1); the theorem is stated for stationary-scale exploration,  $S_T = \Theta(T)$ . (ii)  
 942  $g_1 = \beta |h^* - \psi|$  is  $O(1)$  in general; in the weak-performativity regime it is  $O(\varepsilon)$ , giving the sharper  
 943 factor  $(1 - O(\varepsilon T^{-1/2}))$ . (iii) The fixed- $\delta$  regime is different, and the certainty-equivalent anchor  
 944 is the reason. At fixed prior separation the policy eventually learns  $s$  outright and can exploit it

indefinitely: the exploitation fraction does not vanish (observed directly in V3b’s first, deliberately mis-scaled run). That regime is not a counterexample. Sustained exploitation of learned structure is exactly what D0’s certainty-equivalent anchor re-anchors away: once  $\varepsilon$  is known, the CE-optimal deployment shifts, and the “gain” is no longer measured as a rebate against exploration. The theorem’s content concerns the unlearned residual, whose hardest instances live at the minimax scale  $\delta \sim \sigma/\sqrt{S_T}$ , and there the rebate vanishes at rate  $T^{-1/2}$  (verified). We state this distinction explicitly: it separates the theorem’s content from an apparent counterexample.

**Numerical verification (V3).** Three checks. (V3a) A Bayes-risk simulation with a Gaussian-prior posterior-mean estimator under three designs (front-loaded, spread, adaptive-greedy) with capped budget  $D$ : the measured risk respects  $\sigma^2/(D + \sigma^2/\sigma_\pi^2)$  in every arm, with the spread design near-tight. (V3b-i) The Pinsker chain: the measured posterior imbalance satisfies  $|\mathbb{E}[s | \mathcal{F}_t]| \leq \delta\sqrt{S_t}/\sigma$  pathwise along simulated runs. (V3b-ii) An explicitly exploiting policy, which drifts by the posterior mean of the unknown linear term, simulated at several horizons  $T$ : its realized  $\text{gain}_T/C_T$  decays consistently with  $O(T^{-1/2})$ , and its achieved  $\text{Var}(\hat{\varepsilon}) \times \text{cost}$  never falls below  $\frac{1}{2}\gamma_{\text{PO}}\sigma^2(1 - \text{measured rebate})$ .

**Structure-proofness within reach (T9).** The floors above are nonparametric. A separate threat is a policy that knows the response family’s functional form  $\tau(h) = C_0 + C_1 e^{-ch}$  and shops for designs whose parametric Cramér–Rao product undercuts the floor. Define, for a finitely supported design  $\mu$ ,

$$R(\mu) = (g^\top M(\mu)^{-1}g) \int (h - h^*)^2 d\mu, \quad M(\mu) = \int s s^\top d\mu,$$

with the sensitivity and the estimand gradient

$$s(h) = (1, e^{-ch}, -C_1 h e^{-ch})^\top, \quad g = \nabla_\theta \varepsilon(h^*) = -s'(h^*),$$

normalized so that  $R(\mu) \geq 1$  says the family-knowing product is at least the exchange-rate floor  $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$  under the local-quadratic (A1) cost.

**One-dimensional reduction.** The generalized Rayleigh identity plus the bijection  $u \leftrightarrow \phi_u = u^\top s$  between  $\mathbb{R}^3$  and  $V = \text{span}\{1, e^{-ch}, h e^{-ch}\}$  (using  $C_1 \neq 0$ ) gives

$$g^\top M(\mu)^{-1}g = \frac{1}{\inf \{ \|\phi\|_\mu^2 : \phi \in V, \phi'(h^*) = 1 \}},$$

because  $u^\top g = -\phi'_u(h^*)$  (the sign squares away). So the floor holds at  $\mu$  if and only if some unit-slope family element is  $\mu$ -cheaper than the linear function  $d(h) = h - h^*$ .

**The candidate and its reach.** The unique element of  $V$  with 2-jet  $(0, 1, 0)$  at the anchor is

$$\phi_0(h) = \frac{2}{c} - e^{-c(h-h^*)} \left( \frac{2}{c} + (h - h^*) \right),$$

with expansion  $\phi_0(x) = x - (c^2/6)x^3 + \dots$  in  $x = h - h^*$ . The pointwise domination  $|\phi_0(x)| \leq |x|$  holds exactly on  $x \in [-2/c, \infty)$  (equality only at  $x = 0$  and  $x = -2/c$ ), and fails strictly and exponentially below  $-2/c$ . The three elementary ingredients are  $1 + t < e^t$  (which makes  $\phi_0(x) - x$  strictly decreasing),  $\phi'_0(x) = e^{-cx}(1 + cx) > 0$  above  $-1/c$ , and the single sign change of  $e^{-cx}(1 + cx) + 1$  on  $(-2/c, 0)$  (since  $e^{-cx}(1 + cx)$  is increasing there from  $-e^2$  to 1). The full proof is recorded in the proofs appendix under T9; verified in V9.1.

**Consequence (T9(i)).** For any design supported in  $[h^* - 2/c, \infty)$ :  $R(\mu) \geq 1$  whenever the estimand is estimable, strict as soon as  $\mu$  places mass off the two equality points  $\{h^*, h^* - 2/c\}$ , and  $\inf_\mu R(\mu) = 1$ , approached by symmetric three-point designs collapsing at the anchor (at rate  $\delta^2$  in the collapse half-width  $\delta$ , V9.2; the limit is exact because in 2-jet coordinates the collapsing model is the quadratic family, where every symmetric design has  $R = 1$  on the nose). The trust region A4 with  $r \leq 2/c$  is therefore exactly the condition under which the floor is structure-proof, not a technicality.

983 **Sharpness of the candidate, and an exact duality.** Three facts sharpen the picture, established in  
 984 the adversarial verification campaign with certified numerics (the full write-up is journal material  
 985 under OPEN-5). (i) The easy direction of a duality: for a finite support  $X$  let  $m(X) = \max\{\phi'(h^*) : \phi \in V, |\phi(x)| \leq |x - h^*| \forall x \in X\}$ ; then for every weighting  $w$  of  $X$ ,  $R(X, w) \geq m(X)^2$ , because  
 986 the maximizer  $\psi^*$  scaled to unit slope is feasible in the Rayleigh infimum with  $\int (\psi^*/m)^2 dw \leq$   
 987  $\int d^2 dw/m(X)^2$ . Within reach  $\phi_0$  certifies  $m(X) \geq 1$ , which is exactly T9(i); numerically the duality  
 988 is tight ( $\inf_w R(X, w) = m(X)^2$  by linear-programming duality, certified on random supports), so  
 989 pointwise domination is not a lossy proof device: the reach question *is* the domination question.  
 990 (ii) Single-candidate sharpness: among all unit-slope elements of  $V$ ,  $\phi_0$  is the unique one whose  
 991 domination set contains a full interval  $[h^* - a, h^*)$  with  $a > 0$ , and the maximal such  $a$  is exactly  
 992  $2/c$ . The affine family with 2-jet  $(0, 1, \gamma)$ ,  $\gamma \in (-1, 0)$ , trades the reach boundary outward (to  
 993  $u_2(\gamma) < -2/c$  in units of  $1/c$ ) at the proved price of vacating an interval just below the anchor, and  
 994 the violating designs live exactly in the vacated gap. (iii) The below-reach failure is generic, not a  
 995 boundary artifact: for every  $A > 2/c$  there are nonsingular designs supported in  $[h^* - A, \infty) \cap (0, \infty)$   
 996 with  $R < 1$  (a certified three-point example in the register market attains  $R = 0.8537$ ). Consequence  
 997 for the statement of T9: by the extended-Chebyshev property of  $V$  (any nonzero element has at most  
 998 two zeros counting multiplicity), every design with at least three distinct support points automatically  
 999 has nonsingular information, so the estimability caveat in T9 concerns only one- and two-point  
 1000 designs.  
 1001

1002 **The reach is exact (T9(ii)).** Below  $h^* - 2/c$  the domination fails, and the fail-  
 1003 ure is realized: the frozen witness (support  $\{1.8726, 1.8665, 1.3073, 0.05\}$ , weights  
 1004  $\{0.003329, 0.955702, 0.040697, 0.000272\}$  normalized, frozen anchor  $h^* = 1.8461$ , register  
 1005 market  $c = 1.5$ ) attains

$$R = 0.8516504721831888870 \dots$$

1006 (50-digit precision; V9.2), with every support point except the 0.05 probe inside the reach: one  
 1007 vanishing-weight below-reach probe breaks the floor by 15 percent. Its true-cost ratio is  $1.011 > 1$ ,  
 1008 so the violation lives in the local-quadratic cost model, which is the floor's own scope. Symmetric  
 1009 pair-plus-far-probe families never violate (their ratio tends to 1 from above as the probe weight  
 1010 vanishes; the old scan's verdict of global structure-proofness was a weight-grid artifact, recorded as  
 1011 the eleventh measurement-forced pivot). The characterization of the full below-reach violation set is  
 1012 OPEN-5.

#### 1013 **D4/D5: Design geometry: optimal shapes, dispersion, temporal shaping, Chebyshev support** 1014 **(T5, C5.1, L3, T6)**

1015 This subsection records the derivations behind the three exact design optima of T5, the dispersion  
 1016 corollary C5.1, the temporal-shaping collapse L3, and the Chebyshev counting result T6; it covers  
 1017 both register slots D4 and D5, the Chebyshev material being the D5 portion. All three design  
 1018 optima are derived with proofs, the temporal-shaping lemma and the counting result are proved, and  
 1019 everything is verified in the V4 and V5 numerical checks. It is a derivation record: each optimum is  
 1020 computed in full, the caveats are stated where they arose, and the numerical anchors are those of V4  
 1021 and V5. The formal statements live in the theorem register under the IDs above.

1022 **Setting.** The decision is  $d$ -dimensional and the estimand is the response Jacobian. Exploration has  
 1023 stationary second moment  $M := \mathbb{E}[dd^\top]$  (positive semidefinite), the response noise has variance  $\sigma^2$   
 1024 per coordinate, and by the  $d$ -dimensional form of the cost equivalence lemma L1 (from D0) the value  
 1025 cost rate is

$$\frac{1}{2} \mathbb{E}[d^\top \Gamma_{\text{PO}} d] = \frac{1}{2} \text{tr}(\Gamma_{\text{PO}} M),$$

1026 so the value budget over horizon  $T$  is  $B = (T/2) \text{tr}(\Gamma_{\text{PO}} M)$ . The per-unit-time information matrix  
 1027 is  $M/\sigma^2$ . Absorbing  $T$  and  $\sigma^2$  into units, all three problems take the static normal form of a linear  
 1028 budget constraint  $\text{tr}(\Gamma_{\text{PO}} M) \leq B$  on the design  $M$ .

1029 **A-optimality (T5(a)): what to do if you must know everything equally.** The problem and its  
 1030 exact solution are

$$\begin{aligned} & \text{minimize} \quad \text{tr}(M^{-1}) \quad \text{subject to} \quad \text{tr}(\Gamma_{\text{PO}} M) \leq B, \quad M \succ 0, \\ & M^* = \frac{B}{\text{tr} \Gamma_{\text{PO}}^{1/2}} \Gamma_{\text{PO}}^{-1/2}, \quad \text{tr}((M^*)^{-1}) = \frac{(\text{tr} \Gamma_{\text{PO}}^{1/2})^2}{B}. \end{aligned}$$

1031 Derivation: the objective is strictly convex on the positive definite cone and the constraint is linear, so  
 1032 the KKT conditions are sufficient. Stationarity reads  $-M^{-2} + \lambda \Gamma_{\text{PO}} = 0$ , hence  $M = \lambda^{-1/2} \Gamma_{\text{PO}}^{-1/2}$ ;  
 1033 the budget pins the multiplier through  $\text{tr}(\Gamma_{\text{PO}} M) = \lambda^{-1/2} \text{tr} \Gamma_{\text{PO}}^{1/2} = B$ , so  $\lambda^{-1/2} = B / \text{tr} \Gamma_{\text{PO}}^{1/2}$ ,  
 1034 and substituting back gives the stated value. A diagonal-case Cauchy–Schwarz check confirms the  
 1035 shape: with curvatures  $g_i$  and design weights  $m_i$ ,

$$\left( \sum_i \frac{1}{m_i} \right) \left( \sum_i g_i m_i \right) \geq \left( \sum_i \sqrt{g_i} \right)^2,$$

1036 with equality at  $m_i \propto g_i^{-1/2}$ .

1037 The reading is: explore where the objective is flat, but only to the square-root degree. The optimal  
 1038 exploration covariance is  $\Gamma_{\text{PO}}^{-1/2}$ , the inverse square root and not the inverse: flat directions are cheap  
 1039 per unit information but not infinitely favoured, while curved directions are expensive but cannot be  
 1040 abandoned when every coordinate of the Jacobian is needed.

1041 **The price of isotropic jitter (C5.1).** Isotropic exploration at the same budget,  $M = (B / \text{tr} \Gamma_{\text{PO}}) I$ ,  
 1042 is feasible with the budget tight and has risk  $\text{tr}(M^{-1}) = d \text{tr}(\Gamma_{\text{PO}}) / B$ . The ratio to the A-optimal  
 1043 value is

$$F = \frac{d \text{tr}(\Gamma_{\text{PO}})}{(\text{tr} \Gamma_{\text{PO}}^{1/2})^2} \geq 1,$$

1044 with equality if and only if all curvatures are equal (Cauchy–Schwarz again).  $F$  is the curvature  
 1045 dispersion of the objective: the exact factor by which naive isotropic exploration, which is optimal  
 1046 in LQR-like settings (Simchowitz and Foster, 2020), overpays in the performative setting. This is  
 1047 the content of C5.1, restated as the isotropy contrast P9.2, and  $F$  is computable on the calibrated  
 1048 multi-bond  $\Gamma_{\text{PO}}$  (whose diagonal is given exactly by P9.3).

**D-optimality (T5(b)): what the confidence-volume objective wants.**

$$\begin{aligned} & \text{maximize} \quad \log \det M \quad \text{subject to} \quad \text{tr}(\Gamma_{\text{PO}} M) \leq B, \\ & M^* = \frac{B}{d} \Gamma_{\text{PO}}^{-1}, \quad \log \det M^* = d \log \frac{B}{d} - \log \det \Gamma_{\text{PO}}. \end{aligned}$$

1049 Derivation:  $\nabla \log \det M = M^{-1}$ , so stationarity reads  $M^{-1} = \lambda \Gamma_{\text{PO}}$ ; the objective is concave and  
 1050 the constraint linear, and the budget gives  $\lambda^{-1} = B/d$ .

1051 A caveat recorded at derivation time: the two shapes differ, A-optimality giving  $\Gamma_{\text{PO}}^{-1/2}$  and D-  
 1052 optimality giving  $\Gamma_{\text{PO}}^{-1}$ , and the two must never be conflated in the paper. The algorithm derivation  
 1053 (D6) uses the D-optimal criterion internally (Frank–Wolfe on  $\log \det$ ), while the headline risk  
 1054 statement is A-optimal; both are stated, and the discrepancy factor between them is itself computable  
 1055 and is small when the dispersion  $F$  is moderate.

1056 **c-optimality (T5(c)): what the corrected (PerfGD-style) update actually needs.** This is the new  
 1057 result of the three. The corrected update consumes the response estimate through a single functional,  
 1058 the correction direction  $c$ : in the scalar family the correction  $\Delta$  needs only  $\varepsilon(h^*)$ , and in  $d$  dimensions  
 1059  $c = \nabla_E \Delta$  is a fixed vector once the operating point is known. Estimating the scalar  $c^\top \theta$  alone:

$$\begin{aligned} & \text{minimize} \quad c^\top M^{-1} c \quad \text{subject to} \quad \text{tr}(\Gamma_{\text{PO}} M) \leq B, \\ & M^* = B \frac{c c^\top}{c^\top \Gamma_{\text{PO}} c} \quad (\text{rank one: probe along } c \text{ itself}), \quad c^\top (M^*)^{-1} c = \frac{c^\top \Gamma_{\text{PO}} c}{B}. \end{aligned}$$

1060 Derivation: substitute  $N = \Gamma_{\text{PO}}^{1/2} M \Gamma_{\text{PO}}^{1/2}$ , so that  $\text{tr } N = \text{tr}(\Gamma_{\text{PO}} M) \leq B$ , and  $w = \Gamma_{\text{PO}}^{1/2} c$ ; the  
 1061 objective becomes  $w^\top N^{-1} w$ . By Cauchy–Schwarz,

$$(w^\top w)^2 = (w^\top N^{-1/2} N^{1/2} w)^2 \leq (w^\top N^{-1} w)(w^\top N w),$$

1062 and  $w^\top N w \leq \|w\|^2 \text{tr } N \leq \|w\|^2 B$ , so  $w^\top N^{-1} w \geq \|w\|^2 / B$ , attained at  $N^* = B w w^\top / \|w\|^2$ .  
 1063 Back-substituting,

$$M^* = \Gamma_{\text{PO}}^{-1/2} N^* \Gamma_{\text{PO}}^{-1/2} = B \frac{(\Gamma_{\text{PO}}^{-1/2} w)(\Gamma_{\text{PO}}^{-1/2} w)^\top}{\|w\|^2}, \quad \Gamma_{\text{PO}}^{-1/2} w = c, \quad \|w\|^2 = c^\top \Gamma_{\text{PO}} c,$$

1064 giving  $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$  as stated.

1065 The corrected trio is worth displaying side by side:

$$\text{A-opt: } M^* \propto \Gamma_{\text{PO}}^{-1/2}, \quad \text{D-opt: } M^* \propto \Gamma_{\text{PO}}^{-1}, \quad \text{c-opt: } M^* \propto c c^\top.$$

1066 A-optimality shapes exploration as  $\Gamma_{\text{PO}}^{-1/2}$ , D-optimality as  $\Gamma_{\text{PO}}^{-1}$ , and c-optimality probes along the  
 1067 correction direction itself: the curvature does not tilt the direction at all, it only prices it. The value  
 1068  $c^\top \Gamma_{\text{PO}} c / B$  says that the cost of knowing your correction is the  $\Gamma_{\text{PO}}$ -norm of what the correction  
 1069 asks for. The savings over the A-optimal value is the operational argument for correction-targeted  
 1070 probing: a desk does not need the whole Jacobian, and the geometry says exactly what it may skip.

1071 Two caveats, recorded as they arose. First, an earlier draft of this derivation mis-stated the c-optimal  
 1072 direction as  $\Gamma_{\text{PO}}^{-1} c$ ; the substitution above corrects it. The error is recorded here per the project’s  
 1073 falsification convention, and the corrected direction is locked by the V4 numerical check. Second,  
 1074 rank-one designs are singular (hence the pseudo-inverse in the value); in practice they are regularized  
 1075 by the trust region, and the paper states both the limiting and the regularized forms.

1076 **Temporal shaping is irrelevant (L3): design space collapse. Lemma L3 (temporal shaping**  
 1077 **is irrelevant).** With response noise i.i.d. and independent of the deployment path, the information  
 1078 about the response parameters depends on the exploration process only through its empirical second  
 1079 moment  $\sum_t d_t d_t^\top$ . The computation is one line: the conditional log-likelihood is

$$\sum_t -\frac{(\tau_t - a - \theta^\top d_t)^2}{2\sigma^2},$$

1080 and its Hessian in  $\theta$  is  $-\sum_t d_t d_t^\top / \sigma^2$ , regardless of the temporal order or dependence structure of  
 1081  $\{d_t\}$ . Hence temporally correlated exploration (AR-shaped, cyclic, burst) buys nothing beyond its  
 1082 stationary design measure, and the design problem is exactly the static problems treated above. Scope  
 1083 boundary: the lemma fails when the noise is serially correlated or feedback-coupled (the regime  
 1084 of section 4 of D2), where shaping does matter; this is stated as the boundary of L3 and left to the  
 1085 journal version.

1086 **Chebyshev counting (T6): why curvature is invisible to trajectories.** This is the D5 portion of  
 1087 the record. The structural family of REFLEX theory 1.1 (`structural_response.py`) is

$$\tau(h) = C_0 + C_1 e^{-ch}, \quad \text{parameters } (C_0, C_1, c), \quad p = 3,$$

1088 with sensitivity vector

$$s(h) = (1, e^{-ch}, -C_1 h e^{-ch})^\top.$$

1089 (In this paragraph  $c$  is the decay rate of the structural family, not the correction direction of the  
 1090 c-optimal problem.) The counting argument has four parts.

1091 (i)  $\{1, e^{-ch}, h e^{-ch}\}$  is an extended Chebyshev system on any interval: the relevant Wronskian-type  
 1092 determinants are nonzero (standard).

1093 (ii) Consequently the Fisher matrix  $M(\xi) = \sum_i w_i s(h_i) s(h_i)^\top$  of any design  $\xi$  with fewer than  
 1094 three distinct support points is singular, and the decay rate  $c$  is unidentified. The computation:  
 1095  $\text{rank } M(\xi) \leq \#\text{support}(\xi)$ , so a two-point design gives rank at most  $2 < 3 = p$ , hence  $\det M(\xi) =$   
 1096  $0$  and the  $c$ -direction lies in (or meets) the null space. The exact-arithmetic check in V5 finds

1097  $\det M = 0$  to machine precision for every random two-point design and  $\det M > 0$  for generic  
1098 three-point designs.

1099 (iii) Any noiseless RRM trajectory has effective support at most two (C1.2 of D1: geometric collapse  
1100 for modulus below one, a period-two orbit at the boundary). Hence no retraining trajectory, at any  
1101 modulus, identifies the curvature of its own response family.

1102 (iv) D-optimal designs for this family on a trust-region interval  $[h^* - r, h^* + r]$  are supported on  
1103 exactly three points including both endpoints (a de la Garza-type argument; the interior point is  
1104 computed numerically in the experiments).

1105 This is the sharp form of the repository’s empirical rule that exponential response fits need  
1106 a wide spread range, and the formal reason the anti-echo freeze (the identified flag in  
1107 `structural_response.py`) exists.

1108 **Verified numerically (V4, V5).** Six checks anchor the results above. (1) A-optimality: random  
1109 search over positive semidefinite  $M$  at fixed budget never beats  $(\text{tr } \Gamma_{\text{PO}}^{1/2})^2/B$ , and the closed-form  
1110  $M^*$  achieves it (run at  $d = 5$  with random symmetric positive definite  $\Gamma_{\text{PO}}$ ). (2) D-optimality:  
1111  $M^* = (B/d) \Gamma_{\text{PO}}^{-1}$  beats random feasible designs in  $\log \det$ . (3) c-optimality: random search  
1112 never beats  $c^\top \Gamma_{\text{PO}} c/B$ , and the rank-one construction approaches it. (4) The isotropic factor  
1113  $F = d \text{tr}(\Gamma_{\text{PO}})/(\text{tr } \Gamma_{\text{PO}}^{1/2})^2$  matches the measured ratio. (5) Temporal lemma: AR-shaped and i.i.d.  
1114 exploration with matched empirical second moment give identical OLS covariance. (6) Chebyshev:  
1115  $\det M = 0$  for two-point designs versus  $\det M > 0$  for three-point designs, and the  $c$ -direction  
1116 variance is infinite or huge under two-point designs.

#### 1117 **D6: Safe certainty equivalence (L4, R2, P7.1)**

1118 This derivation records the safety analysis of the certainty-equivalent corrected loop: the structural  
1119 observation R2 (open-loop exploration cannot destabilize the linearized retraining dynamics), the  
1120 perturbed-modulus lemma L4 (how  $C^1$  estimation error in the response moves the closed-loop  
1121 modulus), and the safety proposition P7.1 (the pessimism gate inherits uniform-in-time safety from  
1122 an anytime-valid confidence sequence). The formal statements are recorded in the register under  
1123 these IDs, and the proofs in the proofs appendix. Status of the record: R2 and L4 are derived and  
1124 verified (V6); P7.1 is proved conditional on the validity of the confidence sequence; the  $O(\sqrt{T})$   
1125 design-regret bound in the closing paragraph is stated with a proof strategy only and is labeled a  
1126 journal-version item.

1127 **Where the stability constraint actually bites (R2).** A fact discovered in the derivation and  
1128 promoted to the paper: open-loop exploration cannot destabilize the linearized loop. The linearized  
1129 deviation dynamics under a planned (history-independent) probe sequence  $(u_t)$  read

$$d_{t+1} = -m d_t + u_t,$$

1130 and the homogeneous part keeps its modulus  $m < 1$  regardless of the input  $u$ : bounded inputs give  
1131 bounded outputs. This is R2. The stability risk of probing therefore enters through exactly two  
1132 channels.

1133 First, estimate feedback, the real one. The corrected update deploys

$$h_{t+1} = h_t + \eta \hat{\Phi}'(h_t), \quad \hat{\Phi}' = G + \hat{\Delta},$$

1134 where  $\hat{\Delta}$  is built from the estimated response  $\hat{\varepsilon}(\cdot)$ . Estimation error changes the closed-loop map  
1135 itself; this is where safety must be engineered, and it is the subject of L4.

1136 Second, basin exit, the nonlinear channel: large probes can leave the contraction neighbourhood.  
1137 This is handled by the trust region  $|d_t| \leq r$ , a constraint the design problems of D4 already carry.

1138 **The perturbed-modulus lemma (L4).** Let the corrected map be

$$\Psi(h) = h + \eta [G(h) - \beta(h - \psi) \hat{\varepsilon}(h)],$$

1139 let  $\hat{\rho}$  be its slope at the fixed point, and let  $\rho$  be the slope under the true response  $\varepsilon(\cdot)$ , namely  
1140  $\rho = 1 - \eta \gamma_{\text{PO}}$ , as derived in the REFLEX market model [Nagarajan and Ashok, 2026].

1141 **Lemma L4 (perturbed modulus).** With  $e(h) := \hat{\varepsilon}(h) - \varepsilon(h)$  the  $C^1$  estimation error,

$$|\hat{\rho} - \rho| \leq \eta \beta \left( \|e\|_\infty + |h^* - \psi| \|e'\|_\infty \right) =: \eta \beta L_{\text{corr}} \|e\|_{C^1}.$$

1142 The computation is short. The slope of the corrected map is  $\Psi'(h) = 1 + \eta \hat{\Phi}''(h)$ , so the modulus  
1143 perturbation is  $\eta$  times the curvature error, and

$$\begin{aligned} \hat{\Phi}''(h) - \Phi''(h) &= -\beta \frac{d}{dh} \left[ (h - \psi) e(h) \right] \\ &= -\beta \left[ e(h) + (h - \psi) e'(h) \right]. \end{aligned}$$

1144 Taking absolute values and the supremum over the operating interval gives the bound.

1145 For the structural family the  $C^1$  error is controlled by the parameter error with explicit constants:  
1146 with  $e = \varepsilon(\cdot; \hat{\theta}) - \varepsilon(\cdot; \theta)$ ,

$$\|e\|_{C^1} \leq L_\theta \|\hat{\theta} - \theta\|,$$

1147 where  $L_\theta$  is computable from  $(C_1, c, r)$ ; the constant is stated with the formal proof in the proofs  
1148 appendix. Consequently a confidence region for  $\theta$  translates directly into an interval for the closed-  
1149 loop modulus.

1150 **The pessimism rule and the safety proposition (P7.1).** Maintain any anytime-valid confidence  
1151 sequence  $\Theta_t$  for the response parameters (from the CRN machinery of the REFLEX market model  
1152 [Nagarajan and Ashok, 2026], or a standard self-normalized bound; its width shrinks as information  
1153 accumulates, per D2).

1154 **Rule.** At each step choose the pair (step size  $\eta_t$ , design) so that the worst-case modulus over  $\Theta_t$   
1155 respects the margin,

$$\sup_{\theta \in \Theta_t} |\hat{\rho}(\theta)| \leq 1 - c_{\text{margin}},$$

1156 that is, pessimistic on stability, while the design objective of D4 is evaluated optimistically. When  
1157  $\Theta_t$  is too wide to certify the margin, the rule freezes the correction and explores under the blind  
1158 loop, which is provably stable with modulus  $m < 1$  by R2. This is exactly the anti-echo freeze of  
1159 `structural_response.py`, now derived rather than engineered: this freeze is the pessimism gate.

1160 **Proposition P7.1 (safety).** If the confidence sequence is valid at level  $1 - \delta$  uniformly over time, then  
1161 with probability at least  $1 - \delta$  the realized modulus satisfies  $|\rho_t| \leq 1 - c_{\text{margin}}$  for all  $t$  simultaneously.  
1162 The argument is one line: on the event that  $\theta \in \Theta_t$  for every  $t$ , which has probability at least  $1 - \delta$  by  
1163 anytime-validity, the rule's supremum dominates the realized modulus at every step. The engineering  
1164 reading: safety is inherited from the confidence sequence's uniform validity; there is no union bound  
1165 over steps and no per-step failure accumulation.

1166 **Design regret (stated; journal deliverable).** **Claim (to be proved in full).** The certainty-equivalent  
1167 re-solving scheme (the D-optimal objective from D4, the pessimism rule above, and the trust region  $r$ )  
1168 attains identification regret  $O(\sqrt{T})$  against the budget-matched oracle design. The elliptical-potential  
1169 argument of linear-bandit design theory supplies the rate, and the freeze episodes contribute an  
1170 additive  $O(\log T)$  term: each freeze ends when the confidence width halves, and the widths shrink  
1171 geometrically in the accumulated design energy. As a remark on scope, this item is labeled partial for  
1172 the workshop version: the algorithm and the safety guarantee are proved, while the regret constant  
1173 and the freeze-episode accounting are the journal version's work.

1174 **Numerical verification (V6).** Three checks, all in V6.

- 1175 • Open-loop neutrality: injected bounded probe schedules never change the measured ho-  
1176 mogeneous contraction rate, read off as the slope of the deviation envelope, confirming  
1177 R2.
- 1178 • The perturbed-modulus lemma: corrected loops were run with deliberately corrupted  $\hat{\varepsilon}$ ,  
1179 the  $C^1$  error swept over a grid; in every cell the measured fixed-point slope deviates from  
1180  $1 - \eta \gamma_{\text{PO}}$  by less than the bound of L4, and the bound is tight within a factor of about two  
1181 at small errors.

- The pessimism rule: under a simulated valid confidence sequence, no trajectory in 400 seeded runs violates the margin; under a deliberately invalid (too-narrow) sequence, violations appear, so the validity assumption of P7.1 is necessary, not an artifact of the proof.

## D7: The anchoring crossover (T7)

This subsection records the derivation behind T7: when does anchoring the response estimate to the structural family (“structure”) beat a free-form nonparametric estimator (“capacity”)? The headline of the derivation record is that the crossover is horizon-dependent, not budget-only. The formal statement is recorded as T7 in the theorem register. A status remark carried over from the derivation record: the derivation is complete, including the subtlety that changes the statement of the theorem (the nonparametric bias floor is horizon-dependent), and both the exact secant-bias constant and the two MSE formulas below are verified numerically in V7.

**The question.** REFLEX’s headline empirical finding is that anchoring the response estimate to the structural family beats a free-form estimator: the v3/v4 negative-then-positive result. When is that the right choice *in general*? The naive answer prices misspecification against the exchange rate of T2 alone:

$$\text{anchor} \quad \text{iff} \quad \delta_{\text{mis}}^2 < \frac{\gamma_{\text{PO}} \sigma^2}{2B}.$$

Working the derivation shows the naive answer is incomplete: the nonparametric alternative’s bias is horizon-dependent, and the honest crossover involves  $(\delta_{\text{mis}}, B, T)$  jointly.

**The nonparametric side: local (secant) slope estimation.** Estimate  $\varepsilon = -\tau'(h^*)$  from symmetric probes at  $h^* \pm w$ , using  $n$  probe pairs, under the value budget and the horizon. Writing  $\dot{S} = 2nw^2$  for the design energy, the two constraints read

$$\text{budget:} \quad \frac{1}{2} \gamma_{\text{PO}} S \leq B,$$

$$\text{horizon:} \quad 2n \leq T \implies w^2 = \frac{S}{2n} \geq \frac{S}{T},$$

and with the budget binding,  $S = 2B/\gamma_{\text{PO}}$ , the horizon forces fat probes:

$$w^2 \geq \frac{2B}{\gamma_{\text{PO}} T}.$$

Secant bias, with its exact leading constant. For the symmetric difference quotient,

$$\frac{\tau(h^* + w) - \tau(h^* - w)}{2w} = \tau'(h^*) + \frac{\tau'''(h^*)}{6} w^2 + O(w^4):$$

the even-order terms cancel by symmetry, so the second derivative does not enter at all. Symmetric probing is therefore already bias-optimal at its order, a point worth one sentence in the paper. For the structural family  $\tau(h) = C_0 + C_1 e^{-ch}$  one has  $\tau'''(h) = -c^3 C_1 e^{-ch}$ , so

$$\text{bias}(w) = -\frac{c^3 C_1 e^{-ch^*}}{6} w^2 = -\frac{c^2}{6} \varepsilon w^2,$$

using  $\varepsilon = \varepsilon(h^*) = -\tau'(h^*) = c C_1 e^{-ch^*}$ : a curvature multiple of the slope itself, with the sign fixed by the family (only the magnitude enters the MSE below).

MSE at the budget- and horizon-constrained optimum. Each probe pair yields a difference quotient with variance  $\sigma^2/(2w^2)$ ; averaging the  $n$  pairs gives

$$\text{Var} = \frac{\sigma^2}{2nw^2} = \frac{\sigma^2}{S} = \frac{\gamma_{\text{PO}} \sigma^2}{2B}$$

at the binding budget, independent of how  $S$  splits into  $n$  and  $w$ . The bias, in contrast, requires small  $w$ , and the horizon caps how small:  $w^2 \geq 2B/(\gamma_{\text{PO}} T)$ . Hence, at the constrained optimum,

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left( \frac{\tau'''(h^*)}{6} \right)^2 \left( \frac{2B}{\gamma_{\text{PO}} T} \right)^2.$$

The discovered subtlety, and the reason T7 carries the horizon: the bias term *falls* as  $T$  grows at fixed budget. With more (cheaper, smaller) probes the nonparametric estimator approaches the parametric rate. Nonparametric identification is not uniformly worse; it is worse at *short horizons and fat probes*.

1216 **The parametric (anchored) side.** For a correctly specified family,  $\text{MSE}_p = \kappa_p \gamma_{\text{PO}} \sigma^2 / (2B)$ ,  
 1217 where  $\kappa_p \geq 1$  is the design efficiency for extracting  $\tau'(h^*)$  through the family; c-optimal probing in  
 1218 the sense of T5(c), worked in D4, makes  $\kappa_p \approx 1$ , and the constant is computable. Misspecified by  
 1219  $\delta_{\text{mis}}$ , the  $C^1$  distance from the true response to the family as in D6, the anchored estimator picks up  
 1220 an irreducible squared bias  $\delta_{\text{mis}}^2$ :

$$\text{MSE}_{\text{anchor}} = \kappa_p \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \delta_{\text{mis}}^2.$$

1221 **The crossover.** Anchor iff  $\text{MSE}_{\text{anchor}} < \text{MSE}_{\text{np}}$ , that is,

$$\delta_{\text{mis}}^2 < \left( \frac{\tau'''(h^*)}{6} \right)^2 \left( \frac{2B}{\gamma_{\text{PO}} T} \right)^2 + (1 - \kappa_p) \frac{\gamma_{\text{PO}} \sigma^2}{2B};$$

1222 since  $\kappa_p \geq 1$ , the second term is a (small) penalty on anchoring that vanishes under c-optimal probing  
 1223 through the family. The dominant first term gives the memorable form stated in T7:

$$\text{anchor} \quad \text{iff} \quad \delta_{\text{mis}} < \frac{|\tau'''(h^*)| B}{3 \gamma_{\text{PO}} T} \quad (\text{to leading order}).$$

1224 **Readings.** (i) Anchoring wins at short horizons, tight budgets, and strongly curved responses;  
 1225 the free-form route wins asymptotically in  $T$  at fixed budget. (ii) The formula retrodicts the v3  
 1226 negative result *quantitatively*: the free-form loop failed at modest  $T$  with the trust region forcing fat  
 1227 effective probes, exactly the regime the formula assigns to anchoring. (iii) It also predicts when the  
 1228 finding would reverse (long horizons, generous probe counts), a regime no REFLEX experiment tests  
 1229 [Nagarajan and Ashok, 2026]: a falsifiable novel prediction. (iv) Delta over Lin–Zrníc (literature  
 1230 review item G5): their plug-in analysis has neither the budget nor the horizon; this formula prices  
 1231 both in the paper’s single currency.

1232 **Numerical verification (V7).** Three checks. (1) The secant-bias constant: the measured ratio  
 1233  $(\text{secant} - \tau')/w^2$  converges to  $\tau'''(h^*)/6$  for the exponential family (a deterministic check with  
 1234 exact function evaluations, three values of  $w$  extrapolated). (2) The  $\text{MSE}_{\text{np}}$  formula: Monte-Carlo  
 1235 secant estimation at the constrained optimum matches the two-term formula across a  $(B, T)$  grid.  
 1236 (3) The crossover: simulated anchored fits (correct family plus injected misspecification) against the  
 1237 secant estimator show an empirical crossover  $\delta_{\text{mis}}^*$  tracking the formula within Monte-Carlo tolerance,  
 1238 including its  $B/T$  scaling (halving  $T$  doubles the crossover threshold).

1239 **D8/D9: The ROI of self-knowledge and the LQ and multi-bond instantiations (T8, P9.1, P9.2,**  
 1240 **P9.3)**

1241 This subsection records the derivations behind Theorem T8 (the return on investment of self-  
 1242 knowledge), the two consistency propositions P9.1 and P9.2, and the multi-bond curvature in-  
 1243 stantiation P9.3. Status of the underlying derivation document: the ROI closed forms are derived and  
 1244 verified (V8); both consistency propositions are proved (one-line proofs, verified); the multi-bond  
 1245  $\Gamma_{\text{PO}}$  is derived exactly in the separable case, with the coupled extension stated to first order. The  
 1246 formal statements live in the theorem register and are quoted here only as needed for the algebra.

1247 **Ingredients.** Three constants, all computable from earlier derivations and the primitives of the  
 1248 base structural market theory (the REFLEX market model [Nagarajan and Ashok, 2026]), enter the  
 1249 decision problem:

$$A = \frac{1}{2} \gamma_{\text{PO}} (h_{\text{SP}} - h_{\text{PO}})^2, \quad b = \frac{1}{2} \gamma_{\text{PO}} \sigma^2, \quad a = \frac{1}{2} \gamma_{\text{PO}} \kappa^2.$$

1250 Here  $A$  is the echo-chamber value gap per step (the per-step value of running the corrected deployment  
 1251  $h_{\text{PO}}$  instead of the naive  $h_{\text{SP}}$ , the REFLEX market model [Nagarajan and Ashok, 2026], eq. (6b));  
 1252  $b$  is the exchange rate of T2, under which estimator accuracy  $v$  costs  $b/v$  in performative value  
 1253 (derivation D2); and  $a$  is the residual gap per unit of estimator variance: an error in  $\hat{\varepsilon}$  of variance  $v$   
 1254 displaces the corrected deployment by  $\kappa^2 v$  in squared spread units, where  $\kappa = |dh_{\text{PO}}/d\varepsilon|$  is the  
 1255 fixed-point sensitivity of the corrected optimum to the response slope, available in closed form from  
 1256 the fixed-point equation of the REFLEX market model [Nagarajan and Ashok, 2026].

1257 **The decision problem.** Explore once to accuracy  $v$  (paying the exchange-rate price  $b/v$  up front),  
 1258 then run the corrected deployment forever, discounting the perpetual stream at rate  $\rho_{\text{disc}}$ . The  
 1259 corrected stream is worth  $A - av$  per step, so the net present value is

$$\text{NPV}(v) = \frac{A - av}{\rho_{\text{disc}}} - \frac{b}{v}, \quad v \in (0, \infty).$$

1260 **Optimal accuracy and break-even patience (T8).** Differentiating,

$$\text{NPV}'(v) = -\frac{a}{\rho_{\text{disc}}} + \frac{b}{v^2} = 0 \quad \implies \quad v^* = \sqrt{\frac{b \rho_{\text{disc}}}{a}} = \frac{\sigma}{\kappa} \sqrt{\rho_{\text{disc}}},$$

1261 where the second form follows by substituting  $a$  and  $b$ : the common factor  $\frac{1}{2}\gamma_{\text{PO}}$  cancels, so the  
 1262 optimal accuracy does not involve the curvature at all. Since  $\text{NPV}''(v) = -2b/v^3 < 0$  on  $(0, \infty)$ ,  
 1263 the objective is concave there and  $v^*$  is the global maximizer. At the optimum the two  $v$ -dependent  
 1264 terms are equal,

$$\frac{a v^*}{\rho_{\text{disc}}} = \sqrt{\frac{a b}{\rho_{\text{disc}}}} = \frac{b}{v^*},$$

1265 so the optimal value is

$$\text{NPV}(v^*) = \frac{A}{\rho_{\text{disc}}} - 2 \sqrt{\frac{a b}{\rho_{\text{disc}}}}.$$

1266 Exploring at all is worthwhile iff  $\text{NPV}(v^*) > 0$ , which rearranges to

$$A > 2 \sqrt{a b \rho_{\text{disc}}} \quad \iff \quad \rho_{\text{disc}} < \rho^* := \frac{A^2}{4 a b} = \frac{(h_{\text{SP}} - h_{\text{PO}})^4}{4 \kappa^2 \sigma^2}.$$

1267 The last equality is the substitution  $A^2/(4ab) = (\frac{1}{2}\gamma_{\text{PO}})^2 (h_{\text{SP}} - h_{\text{PO}})^4 / (4 \cdot \frac{1}{2}\gamma_{\text{PO}}\kappa^2 \cdot \frac{1}{2}\gamma_{\text{PO}}\sigma^2)$ ,  
 1268 in which  $\gamma_{\text{PO}}$  cancels completely.

1269 Three readings of the closed forms. (i) The break-even patience  $\rho^* = (h_{\text{SP}} - h_{\text{PO}})^4 / (4\kappa^2\sigma^2)$   
 1270 involves only the deployment gap, the correction's sensitivity, and the response noise: the curvature  
 1271 drops out, a genuinely clean and slightly surprising closed form (it cancels between the gap's value  
 1272 and the exploration price). (ii) Everything on the right-hand side is computable from the primitives of  
 1273 REFLEX theory before exploring, which is the point of the result as a decision rule. (iii)  $v^*$  is the  
 1274 right amount of self-knowledge: more accuracy than  $v^*$  is vanity, less is blindness.

1275 **Frontier consistency (P9.1).** For any deterministic design  $d_t$  with energy  $S_T = \sum_{t \leq T} d_t^2$ , the  
 1276 pathwise exchange-rate product is

$$\frac{\sigma^2}{S_T} \times \frac{1}{2} \gamma_{\text{PO}} S_T = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 :$$

1277 the  $S_T$  cancels, so nothing about the schedule matters. In particular the  $d_t \sim c/\sqrt{t}$  schedules with  
 1278  $S_T = c^2 \log T$ , the Lai–Robbins cost- $O(\log T)$  regime of adaptive design, satisfy the exchange rate  
 1279 of T2 with equality: their celebrated cheapness buys proportionally little information, and they sit  
 1280 exactly on the frontier.

1281 **Isotropy contrast (P9.2).** In the  $d$ -dimensional problem, the ratio of the isotropic design's risk to  
 1282 the A-optimal design's risk at matched budget is the curvature-dispersion factor of D4 and C5.1,

$$F = \frac{d \operatorname{tr}(\Gamma_{\text{PO}})}{(\operatorname{tr} \Gamma_{\text{PO}}^{1/2})^2} \geq 1,$$

1283 with  $F = 1$  iff  $\Gamma_{\text{PO}}$  is a multiple of the identity. That is exactly the LQR-like isotropic case in  
 1284 which naive isotropic exploration is optimal (Simchowitz–Foster 2020); generically  $F > 1$  and  
 1285 grows with the spread of the curvature spectrum. The performative setting is generically anisotropic  
 1286 (the bond-level curvatures  $\gamma_a$  span decades across the calibrated universe), so naive exploration is  
 1287 generically suboptimal: this is the contrast between the two regimes stated in the main text.

1288 **Separable multi-bond curvature (P9.3, exact).** In the multi-bond market with per-bond decisions  
1289  $h_a$  and separable flows, the response Jacobian is diagonal,  $E = \text{diag}(\varepsilon_a)$ , and the performative  
1290 objective separates across bonds. Applying the scalar curvature computation of the REFLEX market  
1291 model [Nagarajan and Ashok, 2026], section 4.1, bond by bond gives, exactly,

$$(\Gamma_{\text{PO}})_{aa} = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a), \quad (\Gamma_{\text{PO}})_{ab} = 0 \quad (a \neq b).$$

1292 All quantities on the right are the calibrated per-bond constants of the REFLEX market model  
1293 [Nagarajan and Ashok, 2026] (`per_bond_constants`). Consequently the dispersion factor  $F$  of  
1294 P9.2 is directly computable on the 128-bond calibrated universe; that computed value is the number  
1295 quoted in the paper’s experiments.

1296 **Factor-coupled case (construction, first order).** With factor-coupled responses,  $E$  diagonal plus  
1297 low rank as in REFLEX theory 1.5, the same differentiation at the operating point gives

$$\Gamma_{\text{PO}} = \Gamma + \beta \left[ 2 E_{\text{sym}} + c_t (\text{diag}(\psi) E_{\text{sym}} - H E_{\text{sym}}) \right] + O(\|E_{\text{offdiag}}\|^2),$$

$$E_{\text{sym}} = \frac{1}{2} (E + E^\top), \quad H = \text{diag}(h_a),$$

1298 which is exact when  $E$  is symmetric and is evaluated at the operating point. The low-rank structure  
1299 of  $E$  is inherited by the correction, so  $\Gamma_{\text{PO}}^{-1/2}$  and  $\text{tr} \Gamma_{\text{PO}}^{1/2}$  remain computable in  $O(dk^2)$  via the same  
1300 Woodbury and eigen-split machinery already used in REFLEX theory 1.5: the instantiation adds no  
1301 new computational burden.

1302 Caveat (honesty convention). The paper states the separable case as an exact result (P9.3) and  
1303 the coupled case only as a first-order construction carrying its explicit  $O(\|E_{\text{offdiag}}\|^2)$  error term,  
1304 matching the reporting conventions of Nagarajan and Ashok [2026].

1305 **Numerical verification (V8).** Five checks. (1)  $v^* = \sqrt{b \rho_{\text{disc}}/a}$  and the  $\text{NPV}(v^*)$  formula match  
1306 a direct numerical scan over  $v$  on three parameter sets. (2) Root-finding  $\text{NPV}(v^*)(\rho_{\text{disc}}) = 0$  in  $\rho_{\text{disc}}$   
1307 recovers  $\rho^* = (h_{\text{SP}} - h_{\text{PO}})^4 / (4\kappa^2 \sigma^2)$  to  $10^{-10}$ , confirming the  $\gamma_{\text{PO}}$  cancellation. (3) The frontier  
1308 proposition:  $c/\sqrt{t}$  schedules measure exactly on-frontier. (4) The dispersion factor on a synthetic  
1309 curvature spectrum spanning two decades:  $F$  matches the measured isotropic-to-optimal risk ratio  
1310 (cross-checked against V4). (5) The separable  $\Gamma_{\text{PO}}$ : a finite-difference Hessian of the separable  
1311 objective matches the per-bond formula entrywise.

## 1312 D Selected verification rows

Table 1: Selected verification rows (full table, seeds, and code in the repository).

| Claim (register)                                           | Measured                 | Predicted                   |       |
|------------------------------------------------------------|--------------------------|-----------------------------|-------|
| Exchange rate, $m=0.3$ , two amplitudes (T2)               | 0.01062, 0.01070         | 0.01056                     | pass  |
| Exchange rate, $m=0.6$ , two amplitudes (T2)               | 0.00549, 0.00569         | 0.00535                     | pass  |
| Saturating environment (outside A1)                        | +16% over rate           | —                           | drift |
| Feedback floor, $m=0.3/0.6/0.85$ (T1 corollary; E1)        | within 0.2–6.4%          | $(\sigma/\gamma)^2/(1-m^2)$ | pass  |
| Design arm $\text{sd}(\hat{c})$ ; jitter/design ratio (T6) | 0.234; $7.8\times$       | Fisher 0.248; $\geq 3$      | pass  |
| SafeD reaches $h_{\text{PO}}$ ; trust region (L4)          | 1.258; 0 violations      | 1.296; 0                    | pass  |
| Isotropic/A-optimal risk ratio, $d=8$ (T5)                 | 1.529                    | $F = 1.506$                 | pass  |
| Nonparametric MSE at $(B, T)$ optimum (T7)                 | 0.0132                   | 0.0123                      | pass  |
| Pricing domain, 4 cells (T2, exact: LQ); agent             | 0.384, 0.192; 12.65      | 0.384, 0.192; 12.5          | pass  |
| Multi-bond shaped/iso steady-state risk (T5)               | 1.20 (flat control 1.00) | $> 1.15$ ( $\approx 1$ )    | pass  |
| Within-reach min CR ratio; witness beyond (T9)             | 1.0005; 0.8517           | $\geq 1$ ; $< 1$            | pass  |

## 1313 E Additional figures

## 1314 F Reproducibility

1315 All results are deterministic from seeds, numpy-only, CPU-only; reported final errors are 300-step  
1316 tail averages over the deployed path, stable within 0.01 under horizon perturbation (worst measured

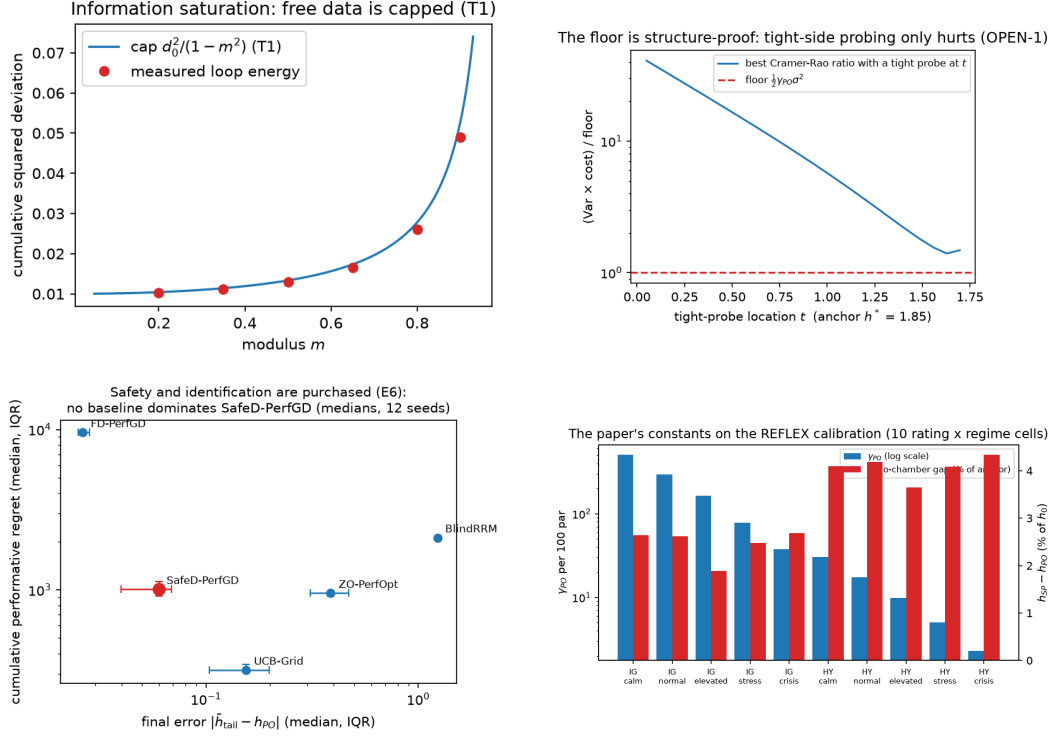


Figure 2: **Top left:** information saturation: measured loop energy against the cap  $d_0^2/(1-m^2)$  (T1). **Top right:** best Cramér–Rao ratio with one tight probe at  $t$  over the coarse weight grid ( $w \geq 0.05$ ): at appreciable weight, tight probes only hurt. The below-reach violation of Theorem 3(ii) lives outside this family (a vanishing-weight probe,  $3 \times 10^{-4}$ , inside an asymmetric cluster), which is exactly why the coarse scan missed it. **Bottom left:** baseline Pareto frontier: no baseline dominates SafeD-PerfGD on median (final error, regret), 12 seeds. **Bottom right:** the paper’s constants on the REFLEX calibration, 10 rating  $\times$  regime cells.

1317 spread 0.0062; the repository’s stability report audits every cell at five horizons). The repository  
 1318 contains: the derivation documents and theorem register (26 results; 38 numerical derivation checks),  
 1319 the pipeline (environments, estimators, designs, agents, baselines; 36 measured-vs-predicted rows;  
 1320 9 unit tests), the real-data leg with its 10/10 port validation, and continuous integration running the  
 1321 verification suites on every push. The register of eleven measurement-forced pivots, each recorded in  
 1322 the code where it happened, with the failed prediction and the fix, is in the repository README.